

Why Trade Over the Counter?*

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Abstract

Over-the-counter trading thrives despite opposing regulatory pressure. In our model, traders who pose a lower adverse selection risk optimally choose the OTC market, where a dealer can offer a trader-specific discount. Closing the OTC market forces some traders to exit but induces others with larger gains from trade to enter. Overall, trade volume falls, average bid-ask spread widens, and yet welfare rises if the asset is *mostly* OTC-traded. An easy-to-implement Pigouvian tax strictly improves welfare over closing the OTC market. We predict that the exchange's spread is wider than the OTC spread and positively correlated with its market share.

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1 Introduction

Over-the-counter (OTC) markets dominate the trading of most financial assets, despite being opaque and difficult to access.¹ This dominance has perplexed policymakers who have sought to shift trades onto transparent and easily accessible exchanges for the past decade. On the forefront of their efforts are the Dodd-Frank Act and the EU’s MiFID II, against which the OTC market has shown remarkable resilience. Under the Dodd-Frank Act, traders of standardized swaps must be offered the option to trade the swaps on an exchange. Nonetheless, 95% of trades in standardized swaps remain over the counter (Nagel, 2016). MiFID II imposes several restrictions on OTC trading, yet the OTC share of trades consistently remained near 20% in EU equities from its 2018 implementation to 2022 (Exton and Healey, 2022).² What explains the resilience of OTC trading? Should the OTC market be closed in the first place? If so, for which assets? We show that the equilibrium outcome deviates starkly from the efficient outcome: closing the OTC market would raise welfare precisely for the *most* OTC-traded assets. Further, our model generates predictions consistent with two empirical observations: (1) The effective spread is lower over the counter than on exchanges; and (2) the share of trades on exchanges is increasing in the exchanges’ quoted spreads.

Our explanation for the resilience of OTC trading builds on its fundamental feature: dealers’ ability to price discriminate among their customers. Price quotes on an exchange are

¹ In our context, “over-the-counter (OTC) markets” consist of all financial markets in which trades are executed non-anonymously between a client and a dealer. This definition includes traditional voice markets in which clients contact dealers one-by-one, and request-for-quote markets in which clients contact multiple dealers at a time. “Exchanges” include all other markets, including limit order books (e.g., stock exchanges), batch auctions, dark pools, and all-to-all request-for-quote platforms.

Limit order books are available for most important asset classes that are nevertheless OTC-dominated (Group of Thirty Working Group on Treasury Market Liquidity, 2021). Examples of such limit order books include NYSE Bonds for corporate bonds, Saxo Bank SaxoTrader for EU government bonds, Tradeweb Dealerweb for repos, Refinitiv FXall for foreign exchange, and Bloomberg SEF CLOB for swaps. These platforms are open to all buy-side trading firms. Beyond these, other platforms (e.g., MarketAxess) feature an anonymous trading option (called Open Trading on MarketAxess). Our model and results do not apply to the illiquid assets that cannot be traded non-anonymously.

²For example, MiFID II bans OTC dealers from matching client orders or rerouting them to brokers or exchanges under most circumstances.

publicly available to all traders. Over the counter, the dealers can and do offer a discount to traders who pose a low risk of adverse selection (say, an insurer rather than a hedge fund), cream skimming them into the OTC market. Neither expanding access to exchanges nor restrictions on dealers that fail to address price discrimination would shift these traders onto exchanges.

We add other key elements of financial markets to analyze social welfare. Traders differ in their gains from trade, and dealers cannot perfectly distinguish an informed trader from an uninformed one—sometimes, hedge funds have liquidity needs and insurers trade on proprietary information. Because heterogeneous gains from trade and imperfect information crucially affect whether and where a trade occurs, we incorporate them into our analysis.

In our model, a continuum of traders choose one of two venues to buy or sell an asset with an uncertain payoff. Uninformed traders have heterogeneous hedging benefits that incentivize them to trade. Informed traders receive imperfect signals about the asset payoff and seek profit. Whether a trader is informed or not is the trader’s private information, which is imperfectly indicated by her public label of either *Likely Informed (LI traders)* or *Likely Uninformed (LU traders)*. All traders optimally choose between trading on an exchange, with a dealer over the counter, or exit. The venues differ solely in that the dealer can condition his prices on each trader’s label. In equilibrium, the LI traders endogenously choose the exchange and the LU traders choose the OTC market.

Trade volumes and bid-ask spreads *should not* guide policy according to our model. Closing the OTC market can raise utilitarian welfare even as it reduces aggregate trade volume and widens average bid-ask spread. This dichotomy between the OTC market’s effects on welfare and trade volume can be stark. Under our baseline model, closing the OTC market always reduces volume, whereas it raises welfare if the share of informed traders is below a single cutoff.

A *cheap substitution* in hedging benefit drives the mismatch between welfare and aggregate volume: The uninformed traders who are induced to trade upon closing the OTC market (“entrants”) substitute for the comparably “cheap” traders who are pushed out (“exiters”). Without the OTC market, the LU traders trade on the exchange and lower its spread down to the No-OTC-market spread S_N . The lower spread induces some uninformed LI traders,

who otherwise would not trade, to do so. Meanwhile, the No-OTC spread S_N exceeds the spread the uninformed LU traders would have paid over the counter, pushing some to exit. Thus, any entrant’s hedging benefit must be above the No-OTC spread S_N , and any exiter’s hedging benefit must be below S_N —the exiters are comparably “cheap”, as they hold the smaller hedging benefits. Aggregate trade volume falls whenever the exiters outnumber the entrants. Cheap substitution overturns this negative effect on welfare if the average entrant has a sufficiently larger hedging benefit than the average exiter.

Precisely, cheap substitution is pronounced if few traders are informed and vanishes with many informed traders. The informed traders absorb a portion of the loss they impose on the dealer and the market maker by paying bid-ask spreads. With relatively few informed traders, the absorbed portion is small. To compensate, the spreads become highly elastic with respect to the loss imposed by the informed traders. That makes the spread of the Likely Informed traders much higher relative to the spread of the Likely Uninformed traders, and cheap substitution is strong. The spread becomes inelastic with many informed traders, and cheap substitution vanishes.

The overall welfare effect of closing the OTC market trades off cheap substitution against a decline in trade volume. Volume declines because the equilibrium spread is less elastic with more informed traders. The spread is thus concave in the mass of informed traders. The concavity pushes the OTC spread farther below the No-OTC spread S_N than the exchange spread is above S_N . This asymmetry lowers aggregate trade volume upon closing the OTC market.

It is unnecessary to directly measure the proportion of informed traders to enact policy. Closing the OTC market raises welfare if the asset is *mostly OTC traded*. We find an analogous result once the traders’ labels become less accurate, perhaps due to laxer disclosure rules. The result follows from (1) the OTC market share is high if most traders are uninformed, in which case (2) closing the OTC market raises welfare. Underlying point (1) is that only the Likely Uninformed traders trade over the counter and there are more LU traders when more traders are uninformed. Point (2) is because cheap substitution is strong with few informed traders and dominates any fall in trade volume upon closing the OTC market. The opposite is true with many informed traders, as cheap substitution vanishes.

We devise an easily implementable Pigouvian tax on OTC trades that attains a strictly higher welfare than closing the OTC market. The Pigouvian tax optimally trades off cheap substitution and the effect of trade volume on welfare. Unlike with closing the OTC market, the tax does not require policymakers to measure informed traders nor determine whether the OTC market share is sufficiently low. Instead a simple experiment suffices: First, one marginally changes the tax t on OTC trades. Second, one computes the Weighted Spread Ratio $|S_E/S_O \div (dV_O/dV_E)|$ (WSR). The ratio of bid-ask spreads S_E/S_O captures the cheap substitution effect, and the ratio of changes in trade volumes $|dV_O/dV_E|$ captures the volume effect. If the WSR is greater than one, cheap substitution dominates and the tax t is below the optimal level t^* . If the WSR is below one, the volume effect dominates and $t > t^*$.

Our model generates two testable predictions. First, cream skimming implies that the OTC spread is lower than the exchange spread. We present supporting evidence in [Internet Appendix II](#). Second, the market share of the exchange and its quoted spread are positively correlated. Adding informed traders simultaneously widens the exchange spread and causes relatively more trades to occur on the exchange. Thus, the quoted spread on the exchange and its market share are correlated through the share of informed traders. [Internet Appendix II](#) documents a positive correlation between the market share of exchanges and their quoted spreads among US equities.

Whether the OTC market is socially beneficial has become an increasingly relevant question as unprecedented regulations are placed on OTC trading. Going beyond Dodd-Frank and MiFID II, the Commodities and Futures Trading Commission banned “name give-up” in the swaps market in 2020, partly to boost trading on exchanges. Most recently, proposals to implement blockchain for financial transactions would reveal traders’ identities to selected dealers. Our results speak to the welfare implications of these plans and policies.

We make three contributions to the literature discussed shortly. First, we introduce cheap substitution, a new mechanism that can overturn the effects of aggregate volume and average spread on welfare. Cheap substitution shows how the two common proxies for welfare—volume and spread—are in fact poor indicators of welfare. Second, we provide novel policy guidance, namely that closing the OTC market improves welfare if the asset is mostly traded over the counter. Third, we introduce an optimal Pigouvian tax that is easy

to implement with local experiments.

1.1 Cheap Substitution, a New Mechanism?

We find that allowing non-anonymous trade can harm social welfare, via cheap substitution. That market segmentation, in our case by anonymity, can be inefficient is well-known. One might rightly wonder if “cheap substitution” merely repackages a known mechanism.

The Market for Lemons

The Lemon’s Problem ([Akerlof, 1970](#)) is a classic mechanism where adverse selection can cause trade to break down. The insight easily extends to how segmenting a functional market could reduce welfare: Select segments in which adverse selection is concentrated may break down, thereby ceasing efficient trades and lowering welfare. Under the Lemon’s Problem, welfare and aggregate volume move in lock-step. Cheap substitution decouples welfare from aggregate volume. It says each trade lost embodies a greater private value than each trade gained. Hence, cheap substitution captures the fall in the *quality* of trades, whereas the Lemon’s Problem captures the decline in the *quantity* of trades.

How do we know cheap substitution drives our results? Whenever we turn off cheap substitution in our model, market segmentation strictly increases welfare and aggregate volume. We devise a Pigouvian tax to set the effective exchange spread equal to the OTC spread, pooling the markets together ([Section 4](#)). By equalizing the private values of the marginal traders in the two markets, cheap substitution is eliminated for any local change. Marginally segmenting the markets by locally cutting the tax strictly increases welfare and aggregate volume. A non-marginal segmentation can lower welfare, despite that it always increases aggregate volume in our base specification. That is, the change in aggregate volume pushes *against* our result that segmentation can reduce welfare.

Binding minimum wage

Price restrictions—in our case, the exchange spread without the OTC market—are known to not always reduce total surplus. [Lee and Saez \(2012\)](#) shows that, where prices and marginal private values are positively correlated (as in our model), a binding minimum wage may have zero effect on the total surplus. In our model, restricting prices to the pooling

spread on the exchange can *strictly raise* the surplus due to a *pecuniary externality* that induces entry. Our restriction reduces the spread of certain traders at the cost of increasing the spread of others, leading to entry in addition to exit. The minimum wage itself only causes exit to unemployment, and a secondary policy change is necessary to increase the total surplus.³ Underlying our results is the private value of the entrants relative to the exiters, as captured by cheap substitution and is absent in the theories of minimum wage.

Externality from OTC markets to exchanges

Pecuniary externalities from OTC markets to exchanges already feature in several models (Rust and Hall, 2003; Desgranges and Foucault, 2005; Bolton, Santos and Scheinkman, 2016; Vogel, 2019). However, pecuniary externalities are pure transfers which, by themselves, do not cause inefficiency. The existing models rely on adding exogenous frictions to the OTC market—search friction (Rust and Hall, 2003; Vogel, 2019), a monopolist dealer (Desgranges and Foucault, 2005), or dealers’ socially wasteful investment and their market power (Bolton et al., 2016)—to derive that OTC trading is generically inefficient.⁴ In the limit where these frictions vanish, allowing OTC trading would no longer be socially harmful in these models. We strip away all exogenous frictions from the OTC market and show how a pecuniary externality *together* with cheap substitution can drive inefficiency. As policies and technologies gradually remove OTC frictions, cheap substitution would remain as a fundamental source of inefficiency from allowing OTC trading.

Non-anonymity in financial markets

The potential of non-anonymity to diminish the payoffs of uninformed traders has been known since Röell (1990). Beneath all ensuing models lay informed traders or large uninformed traders whose orders, once paired with submitters’ identities, are informative about the order book or the asset value.⁵ Therefore, their version of non-anonymity adds a public

³Examples include Allen (1987), Guesnerie and Roberts (1987), and Boadway and Cuff (2001).

⁴In Rust and Hall (2003) and Bolton et al. (2016), allowing OTC trading is always inefficient. In Vogel (2019), it is inefficient whenever the cost to access the exchange is not too high. Desgranges and Foucault (2005) focuses on Pareto efficiency. They show that allowing OTC trading reduces the welfare of traders without a dealer relationship and may reduce that of traders with a relationship.

⁵Examples include Fishman and Longstaff (1992), Foucault, Moinas and Theissen (2007), Rindi (2008), Röell (1990), Admati and Pfleiderer (1991), and Forster and George (1992).

signal about the *aggregate* state of the market, and does not enable price discrimination on individual traders. Non-anonymity can hurt uninformed traders only by worsening prices for all traders, hence the uninformed traders are worse off only if aggregate liquidity falls. In our model, non-anonymity adds signals about the *idiosyncratic* type of each trader without revealing any information about the aggregate state of the market. Price discrimination across such signals causes cheap substitution, which can leave uninformed traders worse off even as it improves aggregate liquidity.

Third degree price discrimination

Our dealer varies his prices across trader characteristics, namely their labels, and therefore engages in *third degree* price discrimination. We show that such price discrimination can lower welfare. [Pigou \(1920\)](#) establishes that a price discriminating monopolist may be worse for social welfare than a non-discriminating monopolist. His argument, now termed the “misallocation effect,” says output is inefficiently distributed whenever different consumers are charged different prices ([Aguirre, Cowan and Vickers, 2010](#)). In their framework, the distribution of private values entirely determines whether price discrimination raises or lowers welfare. Consequently, there can be *no* guidance on the effects of price discrimination on welfare ([Bergemann, Brooks and Morris, 2015](#)).⁶ In our model, adverse selection risk disciplines the effect of price discrimination. The informed traders absorb a larger portion of the dealer’s gross loss as their number increases. With many informed traders, prices become insensitive to traders’ labels, which eliminates cheap substitution. A consequence is our robust guidance on welfare: For *any* distribution of private values, price discrimination raises welfare with many informed traders (where cheap substitution vanishes) and lowers welfare with few informed traders (where cheap substitution is pronounced).

1.2 Related Literature

Most closely related to us are the studies of venue choice between centralized and OTC markets. One strand in this literature abstracts away from adverse selection and focuses

⁶Precisely, price discrimination can generate any outcome under which the producer is at least as well off as with a uniform price, consumers receive non-negative payoffs, and the allocation is possible ([Bergemann et al., 2015](#)).

on the presence of search frictions (Pagano, 1989; Rust and Hall, 2003) or limited trading capacity (Dugast, Üslü and Weill, 2022) in OTC markets. This strand does not explain the resilience of OTC trading despite regulatory pressure. Others, like this paper, feature cream skimming driven by price discrimination (Seppi, 1990; Desgranges and Foucault, 2005).⁷ Seppi (1990) explains why trade sizes are larger over the counter than on exchanges. Desgranges and Foucault (2005) examines endogenous dealer-client relationships as another core feature that distinguishes OTC markets from exchanges. They show that the OTC market is viable if traders are sufficiently likely to be informed (their Proposition 5), which implies a negative correlation between the market share of exchanges and their quoted spreads—the opposite of our prediction. In both models, every trade generates the same per-unit surplus, which rules out cheap substitution.

We belong to the enduring literature that compares centralized and OTC markets. Benveniste, Marcus and Wilhelm (1992), Pagano and Roell (1996), Biais, Foucault and Salanié (1998), Malinova and Park (2013), and Glode and Opp (2019) compare a case with only the exchange against one with just the OTC market. Without endogenous venue choice, this literature cannot study the effects of a tax on OTC trades nor of removing OTC trading as an option.

Section 2 describes the model and derives its unique equilibrium and empirical predictions. Section 3 analyzes utilitarian welfare, aggregate trade volume, and quoted spread (A) upon closing the OTC market and (B) as the traders’ labels become less accurate. Section 4 introduces an easy-to-implement Pigouvian tax that strictly improves welfare relative to closing the OTC market. Section 5 discusses the policy implications of our results. Section 6 concludes with a discussion of some potentially important effects that are not captured by our model.

⁷We are more distantly related to the literature on how traders choose between separate exchanges in the presence of adverse selection (Hendershott and Mendelson, 2000; Zhu, 2014; Pagnotta and Philippon, 2018; Lee, 2019; Chao, Yao and Ye, 2019; Babus and Parlato, 2021; Baldauf and Mollner, 2021). These models do not feature price discrimination, as all traders in the same market faces the same price.

2 A Model of Venue Choice

We first set up the model, which extends [Glosten and Milgrom \(1985\)](#) with endogenous venue choice and imperfect labels for trader types. We then discuss our assumptions in [Section 2.2](#) and derive the unique equilibrium in [Section 2.3](#).

2.1 Trading Game

A dealer, a market maker, a mass μ of informed traders, and a mass 1 of uninformed traders, all risk-neutral, trade an indivisible asset in a three-stage game. A trader buys or sells 1 unit, or exits without trading. The dealer acts as the counterparty to the traders and absorbs net demand in an OTC market. The market maker does so on the exchange. The asset is equally likely to pay $v = 1$ or -1 in the last stage.

An informed trader has a private binary signal which equals the true value v with probability $\alpha \in (1/2, 1)$ and $-v$ otherwise. We call probability α the accuracy of the informed traders' signals. Each uninformed trader is equally likely to be a buyer or a seller, and obtains a hedging benefit b_i upon trading in her desired direction. The hedging benefits are independently and uniformly distributed over $[0, 1]$, $b_i \stackrel{\text{iid}}{\sim} \mathbb{U}[0, 1]$. Whether an uninformed trader is a buyer or a seller and her realized hedging benefit b_i are her private information.

An informed trader is labeled $\ell_i = \text{LI}$ ("Likely Informed") with probability θ and LU ("Likely Uninformed") otherwise. An uninformed trader is LI with probability $1 - \gamma$ and LU otherwise. The labels are informative, $\theta > 1 - \gamma$, in that an LI trader is more likely to be informed than an LU trader. The labels become more informative as their accuracy, θ or γ , increases. We assume probabilities $\theta < 1$ and $\gamma < 1$, so that the labels are imperfectly informative.

In Stage 1, the dealer posts a bid to buy and an ask to sell one unit of the asset to every trader i . The dealer's price can depend on the trader's label $\ell_i \in \{\text{LI}, \text{LU}\}$. The market maker posts one bid and one ask, available to all traders. Prices are competitive in each market: The dealer offers the highest bid and the lowest ask to earn a zero expected profit conditional on the label ℓ_i , while the market maker does so unconditionally. That is, the

OTC market differs from the exchange in one way, that the dealer observes the label ℓ_i before setting the prices for trader i .

In Stage 2, every trader observes all prices, and then makes two decisions: *whether* to buy, sell, or exit, and *where* to trade. [Figure 1](#) summarizes the timing of the model. All distributions, parameters, and the structure of the game are common knowledge.

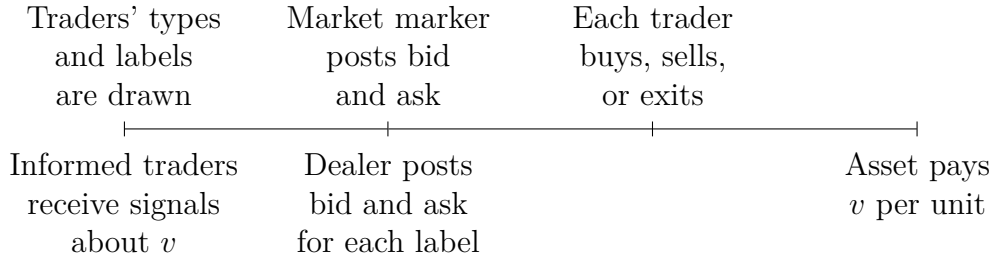


Figure 1: Timeline

We impose a tie-breaking rule to pin down a unique equilibrium.

Assumption 1. *If a trader is indifferent between trading over the counter or on the exchange, she chooses to trade on the exchange.*

[Assumption 1](#) is purely expositional.⁸ The rule is equivalent to imposing a small cost on OTC trades, which can represent the inconvenience of soliciting prices that is absent on exchanges.

2.2 Discussion

Our setup features competitive prices, as defined by [Glosten and Milgrom \(1985\)](#). That prices on exchanges are competitive is a good proxy of reality. However, search friction and dealer market power make OTC prices far from competitive in practice. We assume competitive prices without search friction to show that, despite making the OTC market artificially efficient, closing the OTC market can nonetheless raise welfare. Granting market power to the dealer would further increase the social benefit of closing the OTC market.

⁸All equilibria that would exist without [Assumption 1](#) are payoff equivalent. We can eliminate [Assumption 1](#) and show that traders indifferent between trading on the exchange or in the OTC market must choose the exchange with the same probability ρ . Our results only require that $\rho > 0$.

The dealer in the model posts label-dependent prices. In practice, a trader approaches dealers with a request for quote (RFQ), and the dealers quote trader-specific prices upon receiving the RFQ. We explicitly incorporate RFQs and competition among multiple dealers and market makers in [Internet Appendix I](#). Every trader submits an RFQ to all dealers in the unique extensive-form trembling-hand perfect equilibrium ([Selten, 1975](#)) of the extended model. This equilibrium is otherwise identical to the equilibrium in the next section. We let the dealer post label-dependent prices to simplify exposition.

We interpret a trader’s label as a summary statistic of her observable characteristics and reputation. The observables may include the trader’s industry (hedge fund or insurer), marketing or public filings (active versus passive fund), name (“Two-Sigma” or “AIG”), and any public factoid that is informative about the trader’s motive. The labels are imperfectly informative, as the true motive behind a trade is not known for certain. This uncertainty is illustrated in the commodities futures market, where the US Commodity Futures Trading Commission classifies traders based on their typical strategies. [Cheng and Xiong \(2014\)](#) finds that the trades of “hedgers” often deviate from their label. The hedgers’ trades are far more volatile than output, and the volatility is especially high in their short positions. These positions are consistently profitable and uncorrelated with output, which suggests that even the traders who typically hedge sometimes speculate.

We fix the mass of uninformed traders and will vary the mass of informed traders μ in the welfare analysis. This choice ensures that the maximum welfare that can be achieved is fixed and equal to the total hedging benefit of all uninformed traders.

2.3 Equilibrium

Bid-ask spread characterizes prices in equilibrium. The asset value v is symmetric around zero and the uninformed traders are equally likely to be a buyer or a seller. Thus, an equilibrium bid is the negative of the corresponding ask. Then a sufficient statistic for each pair of bid and ask prices is the half of spread s between those prices. For brevity, we write “half bid-ask spread” and “spread” interchangeably.

The traders’ optimal strategies are simple. An uninformed buyer or seller trades in her

corresponding direction at her best available spread if that spread is less than her hedging benefit. An informed trader buys or sells if her best spread is below the conditional asset value $2\alpha - 1$.

Definition 1. An equilibrium consists of the dealer's spreads for the LI and the LU traders, the market maker's bid-ask spread, and the traders' trading strategies given the available spreads. In equilibrium, each trader's strategy maximizes her expected profit. Given the traders' strategies, the dealer posts the lowest bid-ask spread to earn a zero expected profit conditional on the trader's label, while the market maker does so unconditionally.

Now we derive the zero-profit conditions. If the dealer or the market maker (together "intermediaries") charges a bid-ask spread s , he earns s per trade. Given spread s , an uninformed trader trades with probability $(1 - s)$. Thus, an intermediary's profit is $s \cdot (1 - s)$ per unit mass of uninformed traders who *choose* the intermediary.⁹ However, the intermediary expects a net loss of $(2\alpha - 1 - s)^+$ per trade with an informed trader. Together, the zero-profit condition is

$$\underbrace{s \cdot (1 - s)}_{\text{Profit from uninformed traders}} = \underbrace{(2\alpha - 1 - s)^+ \cdot \beta}_{\text{Loss to informed traders}}, \quad (1)$$

where the *informed ratio* β is the ratio of the mass of informed traders who choose the intermediary's market to the mass of uninformed traders who choose the same. The zero-profit condition (1) has a unique solution:

$$S(\beta) = \frac{1}{2} \left(\beta + 1 - \sqrt{(\beta - 1)^2 + 8(1 - \alpha)\beta} \right).$$

Proposition 0. (a) If the OTC market is closed, the equilibrium spread on the exchange is the No-OTC spread $S_N = S(\mu)$. (b) If the OTC market is open, every LU-trader chooses the OTC market and receives the OTC spread $S_O = S\left(\frac{1-\theta}{\gamma} \mu\right)$ in equilibrium, and every LI-trader chooses the exchange and receives the exchange spread $S_E = S\left(\frac{\theta}{1-\gamma} \mu\right)$.

⁹We say "a trader chooses a market" if the trader would trade in that market were she forced to trade.

Part (a) is a standard result of [Glosten and Milgrom \(1985\)](#). Part (b) incorporates venue choice: The LI traders choose the exchange, and the LU traders choose the OTC market in equilibrium. The traders of the same label choose the same market since, if they split, the prices they face in the two markets must be equal. Then, all traders with that label would choose the exchange by [Assumption 1](#). Hence, in equilibrium, the LU traders choose the OTC market for its lower spread, while the LI traders are indifferent between the two markets (and choose the exchange due to [Assumption 1](#)). Our results remain unchanged if the LI traders choose the exchange with any strictly positive probability.

2.4 Empirical Predictions

The model supplies several testable predictions. First, the assets that mostly attract traders who pose low adverse selection risk (say, insurers and passive funds) have high OTC market shares. Second, trading costs are lower over the counter than on exchanges. Third, the share of trades on exchanges and the exchanges' quoted spreads are positively correlated, driven by variation in informed trading. [Proposition 1](#) formalizes these predictions.

We let V_O denote the equilibrium volume of trades over the counter, V_E the trade volume on the exchange, and $V := V_O + V_E$ the aggregate trade volume.¹⁰ Market shares are V_O/V over the counter and V_E/V on the exchange.

Proposition 1. *Given any triple (α, θ, γ) :*

- (a) *The bid-ask spread is strictly lower over the counter than on the exchange, $S_O < S_E$.*
- (b) *Both the exchange spread S_E and its market share V_E/V are strictly increasing in the informed mass μ .*

Part (a) is a direct consequence of cream skimming. Part (b) rests on the informed traders imposing adverse selection risk and mechanically increasing the ratio of LI to LU traders, which raises exchange market share.

¹⁰Explicitly, $V_O = (1 - \theta)\mu + \gamma \cdot (1 - S_O)$ and $V_E = \theta\mu + (1 - \gamma) \cdot (1 - S_E)$.

We cite evidence related to the predictions of [Proposition 1](#) in [Internet Appendix II](#). We document a positive correlation between exchange market share and exchange quoted spreads among US-listed stocks in [Internet Appendix III.3](#).

3 Welfare, Volume, and Spread

We analyze utilitarian welfare, aggregate trade volume, and average bid-ask spread (A) upon closing the OTC market or (B) as the traders’ labels become less accurate, perhaps due to laxer disclosure requirements. Closing the OTC market or lowering label accuracy raises welfare if the share of trades in the OTC market is *high*, and lowers welfare if the OTC share is *low*. Moreover, welfare can decline while the aggregate volume rises and the average spread falls.

[Section 3.1](#) formally presents our main result. [Section 3.2](#) explains how our results arise from two opposing effects: “cheap substitution” that raises welfare upon (A) or (B) versus “volume effect” that lowers welfare. [Section 3.3](#) shows that our findings are robust when the distribution of the hedging benefits b is generalized to a generic distribution F .

3.1 Main Result

We define *welfare* and *average bid-ask spread* $\bar{S}$ as follows.

Welfare W is the sum of all agents’ ex-ante payoffs. (2)

Average bid-ask spread $\bar{S}$ is the volume-weighted average of the bid-ask spreads (3)
in the OTC market and on the exchange, $\bar{S} := \frac{V_E}{V} S_E + \frac{V_O}{V} S_O$.

In our model, welfare W measures the total gains from trade and equals the total hedging benefits of the uninformed traders who elect to trade. “Lower average spread $\bar{S}$ ” is interchangeable with “higher aggregate volume V ,” because the average spread $\bar{S}$ is the inverse

of the aggregate volume V in equilibrium.¹¹

Proposition 2. *Given any triple (α, θ, γ) and any constants $\theta_0, \theta_1 \in (1 - \gamma, 1)$, $\theta_1 > \theta_0$:*

- (a) *There exists a cutoff $\mu^* > 0$ such that closing the OTC market strictly raises welfare W (defined in (2)) across all mass of informed traders $\mu < \mu^*$ and strictly lowers W across all $\mu > \mu^*$. Equivalently, closing the OTC market strictly raises W if the OTC market share V_O/V is above a single cutoff and strictly lowers W if V_O/V is below the single cutoff.*
- (b) *There exists a cutoff $\hat{\mu} > 0$ such that lowering the traders' label accuracy θ from θ_1 to θ_0 raises W across all $\mu < \hat{\mu}$ and strictly lowers W across all $\mu > \hat{\mu}$. Equivalently, lowering θ strictly raises W if V_O/V is above a single cutoff and strictly lowers W if V_O/V is below the single cutoff.*
- (c) *Closing the OTC market or lowering θ from θ_1 to θ_0 strictly reduces the aggregate trade volume V and widens the average bid-ask spread $\bar{S}$ (defined in (3)) across all $\mu > 0$.*

Proposition 2 provides a simple guide for policy. Closing the OTC market or lowering label accuracy improves utilitarian welfare for assets mostly traded over the counter, and harms welfare for assets mostly traded on exchanges. In stark contrast, conventional measures of liquidity—aggregate trade volume and average bid-ask spread—*always worsen*. Hence, trade volumes and spreads are poor measures of welfare in our model.

Figure 2 illustrates Proposition 2 with parameters $\theta = \gamma = 0.8$ and $\alpha = 0.98$. It plots the numerical changes in welfare W , aggregate trade volume V , and average spread $\bar{S}$ upon closing the OTC market, across OTC market share. We vary the mass of informed traders μ to generate W , V , $\bar{S}$, and the OTC market share. The changes are positive above the red line. Figure 2 confirms that closing the OTC market always reduces trade volume and widens

¹¹Competitive prices imply that the market maker and the dealer's aggregate revenue equals their gross loss from trading with the informed traders in equilibrium. Formally,

$$\underbrace{V \cdot \bar{S}}_{\text{Revenue}} = \underbrace{(2\alpha - 1) \cdot \mu}_{\text{Gross loss}},$$

which yields the inverse relationship between V and $\bar{S}$.

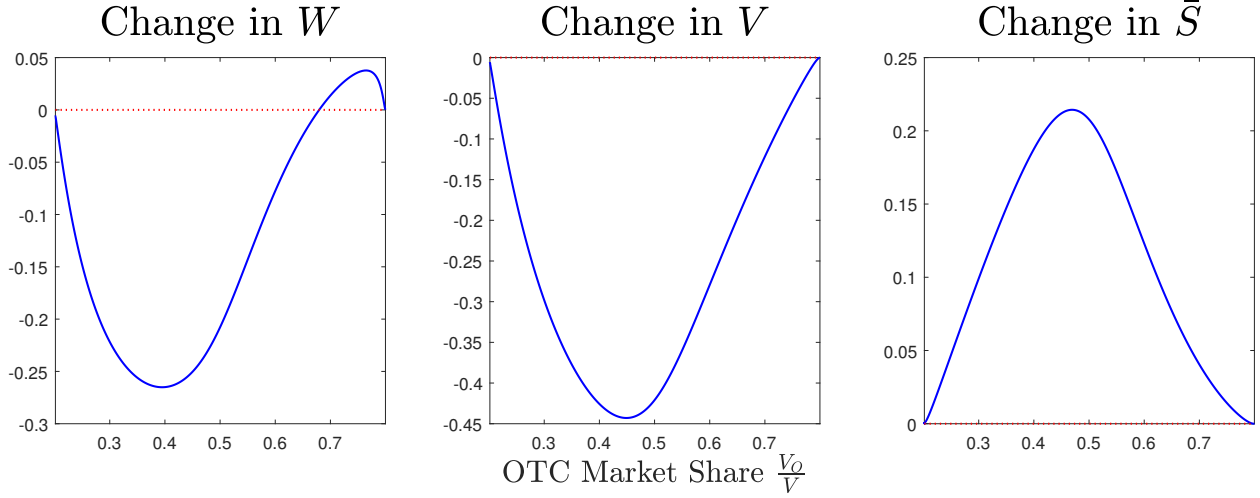


Figure 2: Effects of Closing the OTC Market

average spread, whereas welfare rises if the OTC market share is high and falls if it is low. The rise and the fall in welfare are of the same order of magnitude. Other OTC frictions not modeled here—such as search frictions or dealers’ market power—would contribute to the rise and lessen the fall. Therefore, adding search frictions or relaxing competitive prices in the OTC market would only expand the range of parameters under which closing the OTC market raises welfare.

3.2 Explanation: Cheap Substitution and the Volume Effect

Proposition 2 has two fundamental messages. First, there is a generic mismatch between the effects of policies on traders’ welfare versus aggregate volume or average spread. Second, closing the OTC market or lowering label accuracy improves welfare if the mass of informed traders μ is small, and harms welfare if μ is large. The trade-off between cheap substitution and a volume effect drives our results. Below, we illustrate this trade-off upon closing the OTC market.¹² (Proofs are in [Appendix A](#).)

The volume-welfare mismatch

¹²Closing the OTC market is equivalent to lowering traders’ label accuracy θ from the current level $\theta_1 > 1 - \gamma$ to the uninformative level $\theta_0 = 1 - \gamma$. This is why the two counterfactuals share the same intuition and the same effect on welfare and volume.

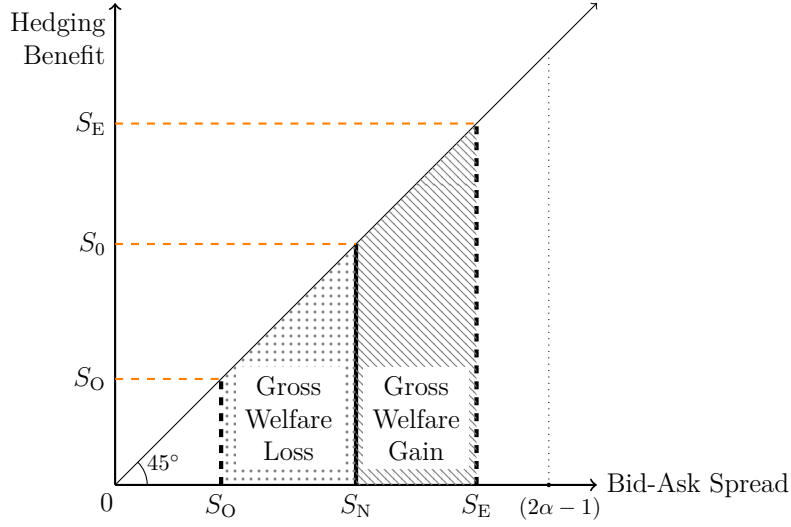


Figure 3: Cheap Substitution

Upon closing the OTC market, some uninformed LU traders exit while some uninformed LI traders enter. The exiters have smaller hedging benefits than the entrants. In welfare terms, the exiters are “cheap” relative to their substitutes.

Cheap substitution drives the mismatch between volume and welfare, as [Figure 3](#) shows. The 45-degree line marks the hedging benefit of a marginal uninformed trader for a given spread. With the OTC market, the spread on the exchange is S_E , and the spread over the counter is S_O . Upon closing the OTC market, some uninformed LU traders exit since their spread rises from the OTC spread S_O to the No-OTC spread S_N . Those exiters’ hedging benefits are in the range $[S_O, S_N]$. Meanwhile, the LI traders’ spread falls from the exchange spread S_E to the No-OTC spread S_N , inducing some uninformed LI traders to enter. These entrants’ hedging benefits are in the strictly higher range $[S_N, S_E]$. In welfare terms, the exiters are “cheap” relative to their substitutes. Cheap substitution places a natural upward pressure on welfare upon closing the OTC market: welfare declines only if the fall in volume is large enough to overcome the positive effect of cheap substitution on welfare.

Cheap substitution versus the volume effect

Why does closing the OTC market raise welfare if and only if the mass of informed traders μ is small? Because cheap substitution is pronounced with a small informed mass μ , exactly where the countervailing *volume effect* is weak. Upon closing the OTC market,

welfare adjusts by the newly attained hedging benefits of the entrants (“gross welfare gain”) less the hedging benefits of the exiters (“gross welfare loss”):

$$\begin{aligned}\text{Gross Welfare Gain} &= \bar{b}(\text{entrants}) \times m(\text{entrants}), \\ \text{Gross Welfare Loss} &= \bar{b}(\text{exitors}) \times m(\text{exitors}).\end{aligned}$$

Terms $\bar{b}(\text{exitors})$ and $\bar{b}(\text{entrants})$ are the exiters’ and the entrants’ average hedging benefits, and $m(\text{exitors})$ and $m(\text{entrants})$ are their respective mass. It is immediate from these definitions that

$$\text{welfare rises if and only if } \underbrace{\frac{\bar{b}(\text{entrants})}{\bar{b}(\text{exitors})}}_{\text{Cheap Substitution}} > \underbrace{\frac{m(\text{exitors})}{m(\text{entrants})}}_{\text{Volume Effect}}. \quad (4)$$

Condition (4) precisely defines cheap substitution and the volume effect. Upon closing the OTC market, each entrant gains a larger hedging benefit than each exiter (cheap substitution), yet the exiters outnumber the entrants (volume effect). Cheap substitution is defined as $\bar{b}(\text{entrants})/\bar{b}(\text{exitors})$ and captures how much the entrants’ hedging benefits outsize those of the exiters. The volume effect is defined as $m(\text{exitors})/m(\text{entrants})$ and measures the relative magnitude by which the exiters outnumber the entrants. We illustrate these terms in [Figure 4](#).

Elasticity of the bid-ask spread determines the strength of cheap substitution and the volume effect. We write $e(\beta)$ to denote the elasticity of the equilibrium spread $S(\beta)$ with respect to the ratio of informed to uninformed traders β . We first take as given that the spread elasticity $e(\beta)$ is below 1 and decreasing in the mass of informed traders μ . We later explain why.

Cheap substitution $\bar{b}(\text{entrants})/\bar{b}(\text{exitors})$ is well above 1 with a small mass of informed traders μ (left panel of [Figure 4](#)) and falls toward 1 with a large informed mass μ (right panel). This pattern holds because the spread elasticity $e(\beta)$ is decreasing in μ . With a small informed mass μ , the equilibrium spread $S(\beta)$ is highly elastic, which pushes up the ratio of the exchange spread to the OTC spread S_E/S_O . The entrants’ hedging benefits,

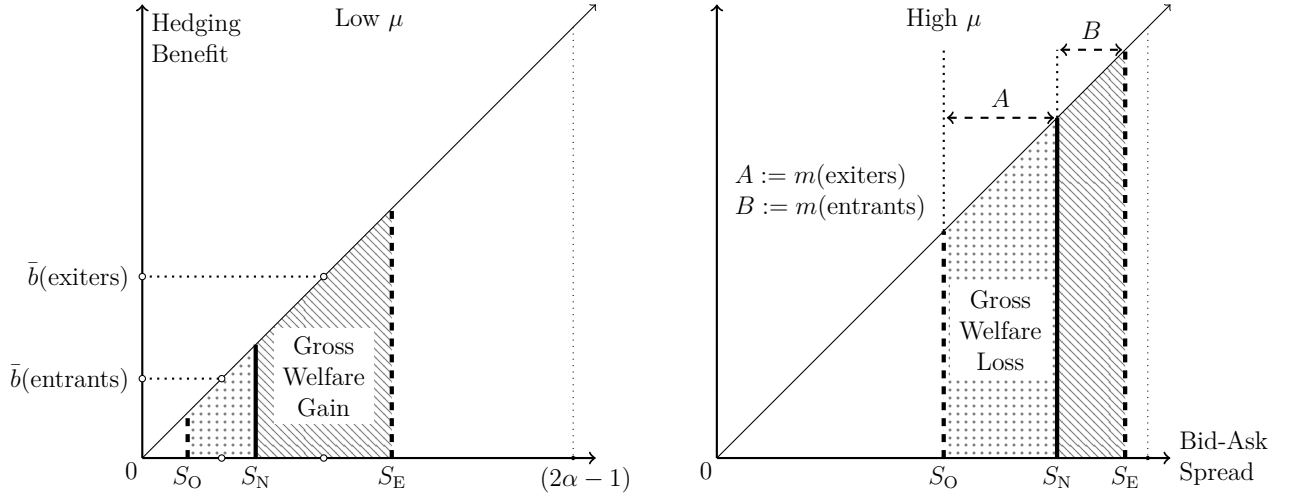


Figure 4: Closing the OTC market improves welfare if μ is small

Cheap substitution dominates if the mass of informed traders μ is small. The volume effect dominates if the informed mass μ is large.

in range $[S_N, S_E]$, are thus far larger relative to the exiters' benefits $[S_O, S_N]$, raising cheap substitution $\bar{b}(\text{entrants})/\bar{b}(\text{exitors})$ well above 1.¹³ With a large informed mass μ , the spread $S(\beta)$ is inelastic and the ratio of the spreads S_E/S_O approaches 1, which pulls down cheap substitution $\bar{b}(\text{entrants})/\bar{b}(\text{exitors})$ towards 1.

The **volume effect** has the opposite pattern as cheap substitution. Part (c) of **Proposition 2** says aggregate trade volume strictly decreases upon closing the OTC market—the exiters *always* outnumber the entrants, $m(\text{exitors})/m(\text{entrants}) > 1$. Moreover, the volume effect $m(\text{exitors})/m(\text{entrants})$ nears 1 with a small informed mass μ and rises well above 1 with a large μ .

Closing the OTC market reduces aggregate volume (**Proposition 2** (c)), because the spread elasticity $e(\beta)$ is below 1. This means the equilibrium spread $S(\beta)$ is *concave*. **Figure 5** illustrates how closing the OTC market widens the LU traders' spread by more than it lowers the LI traders' spread, $S_N - S_O > S_E - S_N$. Therefore, more LU traders exit than the mass

¹³In the limit when $\mu \rightarrow 0$, $\lim_{\mu \rightarrow 0} \bar{b}(\text{entrants})/\bar{b}(\text{exitors}) = \lim_{\mu \rightarrow 0} (S_N + S_E)/(S_N + S_O) = (1 - \gamma + \theta)\gamma/[(\gamma + 1 - \theta)(1 - \gamma)] > 1$.

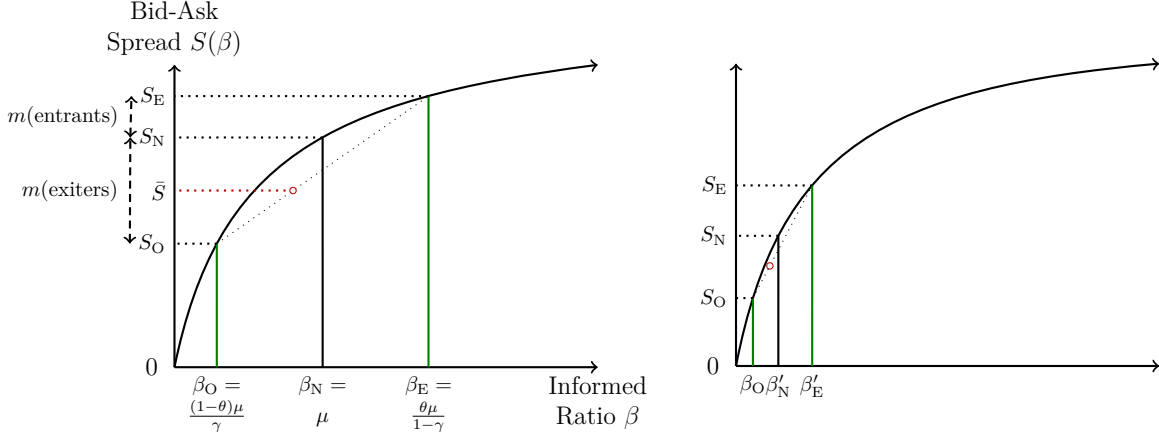


Figure 5: Equilibrium spread is concave ($\mu > \mu'$)

of LI traders who enter, reducing aggregate volume. This volume effect is weak with a small mass of informed traders μ (right panel of [Figure 5](#)). The reason is that the ratios of informed to uninformed traders, β_E and β_O , are close to each other when the informed mass μ is small, and thus the spread curve is locally linear between $[\beta_O, \beta_E]$. The local linearity leaves the OTC spread S_O and the exchange spread S_E equidistant from the No-OTC spread S_N , $S_N - S_O \approx S_E - S_N$, which reduces the volume effect $m(\text{exiters})/m(\text{entrants})$ to 1.¹⁴ When instead the informed mass μ is large, the informed ratios β_E and β_O are far apart, which strengthens the volume effect. In the limit as μ grows unboundedly, the volume effect $m(\text{exiters})/m(\text{entrants})$ remains well above 1.¹⁵

The equilibrium spread $S(\beta)$ is concave, due to that the informed traders, by paying their

¹⁴We can show that the volume effect vanishes when $\mu \rightarrow 0$:

$$\lim_{\mu \rightarrow 0} \frac{m(\text{exiters})}{m(\text{entrants})} = \lim_{\mu \rightarrow 0} \frac{\gamma (S_N - S_O)}{(1 - \gamma) (S_E - S_N)} = 1.$$

¹⁵We can show, when $\mu \rightarrow \infty$, the volume effect

$$\lim_{\mu \rightarrow \infty} \frac{m(\text{exiters})}{m(\text{entrants})} = \frac{\theta \gamma}{(1 - \theta)(1 - \gamma)} > 1.$$

bid-ask spreads, contribute to the dealer's break-even revenue $(2\alpha - 1)\beta$.¹⁶ If the ratio of informed to uninformed traders β doubles, the spread $S(\beta)$ less than doubles, because all the extra informed traders pay the spread. This way, the spread elasticity $e(\beta)$ remains below 1, and the spread $S(\beta)$ is concave. Further, the spread elasticity $e(\beta)$ is decreasing in the informed mass μ . When the informed traders' mass μ rise, their trade volume increases by a larger factor than the uninformed traders' volume. Hence, the total trade volume increases by a factor that is rising in the informed traders' share of the total volume, and this share is higher with a larger informed mass μ . Therefore, when μ is higher, the bid-ask spread only needs to make up for a smaller portion of any change in the break-even revenue $(2\alpha - 1)\beta$, and the spread elasticity $e(\beta)$ is decreasing in μ .

In sum, cheap substitution dominates the volume effect when the mass of informed traders μ is small. Closing the OTC market thereby only raises welfare with relatively few informed traders. [Section 3.3](#) shows that the trade-off between cheap substitution and the volume effect persists when the hedging benefits b follow a general distribution.

3.3 General Distributions

Our welfare implications are remarkably sharp: Closing the OTC market or lowering label accuracy always reduces aggregate volume, while their effect on welfare hinges on a single cutoff μ^* over the mass of informed traders. Crucial for the result is the rate of endogenous entry versus exit, which plainly depends on the distribution of hedging benefits. In this section, we show that our welfare implications are robust to most economically relevant distributions.

To do so, we assume that the hedging benefits follow a generic distribution F , $b_i \stackrel{iid}{\sim} F$, whose support is $[0, 1]$ with a pdf $f(s)$. Such a distribution is “regular” in that its pdf $f(s)$

¹⁶The break-even revenue is the gross value of the informed traders' private information in a market,

$$\underbrace{(2\alpha - 1)}_{\text{Expected value of each signal}} \cdot \underbrace{\beta}_{\text{Ratio of informed to uninformed traders}} ; \quad \beta \in \{\beta_E, \beta_O, \beta_N\}.$$

It is the exogenous part of the zero-profit condition (1).

is differentiable in some neighborhoods of 0 and $2\alpha - 1$ (the lower and upper bounds of the bid-ask spread) and that the limits $\lim_{s \downarrow 0} f'(s)$ and $\lim_{s \uparrow 2\alpha - 1} f'(s)$ exist.¹⁷ With a general distribution F , the zero-profit condition (1) becomes

$$\underbrace{s \cdot (1 - F(s))}_{\text{Profit from uninformed traders}} = \underbrace{(2\alpha - 1 - s)^+ \cdot \beta}_{\text{Loss to informed traders}}. \quad (5)$$

The left-hand side is changed from $s \cdot (1 - s)$ to $s \cdot (1 - F(s))$, while the right-hand side is unchanged.

Two measures fully characterize the effects of closing the OTC market or lowering label accuracy on volume and welfare under this general framework. *Marginal volume* Δ_V is the decrease in the proportion of uninformed traders who trade when the informed ratio β marginally increases,

$$\Delta_V(\beta) := \underbrace{S'(\beta)}_{\text{Increase in spread}} \cdot \underbrace{f(S(\beta))}_{\text{Decrease in trades}}. \quad (6)$$

Marginal welfare Δ_W is the decrease in welfare per unit mass of uninformed traders when the informed ratio β marginally increases. It is equivalent to the marginal volume Δ_V weighted by the hedging benefit of the marginal uninformed trader, $b = S(\beta)$:

$$\Delta_W(\beta) := \underbrace{S(\beta)}_{\text{Benefit lost}} \cdot \underbrace{S'(\beta)f(S(\beta))}_{\Delta_V}. \quad (7)$$

We show in [Appendix A.2](#) that both the marginal volume Δ_V and the marginal welfare Δ_W are well-defined.

The following proposition says our welfare implications hold under any distribution that

¹⁷These limits can be, but need not be, infinite. Most economically relevant pdfs for f are regular. For example, any pdf f continuously differentiable in the support $[0, 1]$ and any beta distribution are regular.

admits an “n-shaped” marginal welfare Δ_W . We sketch an example in [Figure A.3](#).

Marginal welfare Δ_W is *n-shaped* if Δ_W is strictly increasing below a cutoff in the ratio β and strictly decreasing beyond the cutoff. (8)

Proposition 3. *We let F be any regular distribution with support $[0, 1]$ and each uninformed trader’s hedging benefit $b_i \stackrel{iid}{\sim} F$. Given any triple (α, θ, γ) and any constants $\theta_0, \theta_1 \in (1 - \gamma, 1), \theta_1 > \theta_0$:*

- (a) *There exists two cutoffs $\bar{\mu}$ and $\underline{\mu}$, $\bar{\mu} > \underline{\mu} > 0$, such that closing the OTC market strictly raises welfare W across all mass of informed traders $\mu < \underline{\mu}$ and strictly lowers W across all $\mu > \bar{\mu}$.*
- (b) *There exists two cutoffs μ^H and μ^L , $\mu^H > \mu^L > 0$, such that lowering the traders’ label accuracy θ from θ_1 to θ_0 raises W across all $\mu < \mu^L$ and strictly lowers W across all $\mu > \mu^H$.*
- (c) *If the marginal welfare Δ_W is n-shaped (defined in (7)-(8)), the cutoffs $\bar{\mu} = \underline{\mu}$ and the cutoffs $\mu^H = \mu^L$.*
- (d) *If the marginal volume $\Delta_V(\beta)$ (defined in (6)) is decreasing, closing the OTC market or lowering θ from θ_1 to θ_0 strictly reduces the aggregate trade volume V and widens the average bid-ask spread $\bar{S}$ across all $\mu > 0$.*

[Proposition 3](#) parts (a) and (b) say, under *any* regular pdf f , closing the OTC market or lowering label accuracy raises welfare for a small mass of informed traders μ and lowers welfare for a large informed mass μ . Part (c) sharpens our policy guidance to a single cutoff on the informed mass μ whenever marginal welfare Δ_W is n-shaped. [Proposition 4](#) later shows that most relevant distributions for F would result in an n-shaped marginal welfare Δ_W . Part (d) emphasizes how the measures of liquidity are poor indicators of welfare: Under a sufficient condition, closing the OTC market or lowering label accuracy always reduces the aggregate volume and widens the average spread.

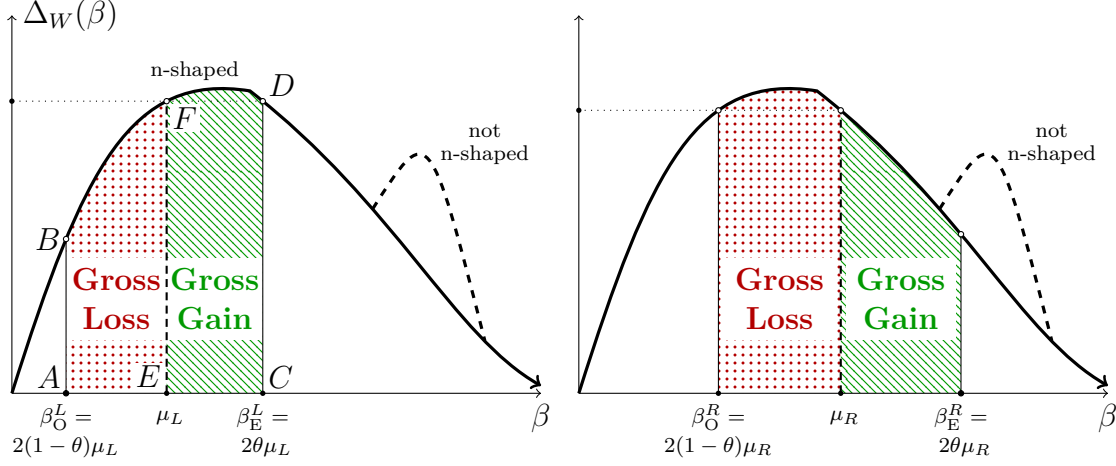


Figure 6: The Effect of Closing the OTC Market on Welfare ($\gamma = 1/2$, $\theta > 1 - \gamma = 1/2$)

We now outline the proof of [Proposition 3](#) parts (a) and (c). (We generalize this proof in [Appendix A.2](#) for part (b).) We consider the simplest example where the label accuracy $\gamma = 1/2$ (and let $\theta > 1 - \gamma = 1/2$) and the marginal welfare Δ_W is continuous. [Figure 6](#) plots the marginal welfare Δ_W upon closing the OTC market in this example. We first focus on the left panel, where the mass of informed traders is μ_L . With the OTC market, the informed ratio is $\beta_O^L = 2(1 - \theta)\mu_L < \mu_L$ over the counter and $\beta_E^L = 2\theta\mu_L > \mu_L$ on the exchange. As the informed ratio without the OTC market $\beta = \mu_L$ is higher than the OTC ratio β_O^L , the LU traders pay a higher spread upon closing the OTC market. The resulting exit by uninformed LU traders creates a gross welfare loss, marked by the red-dotted area between the points $\beta = \mu_L$ and $\beta = \beta_O^L$. Symmetrically, the gross welfare gain from the entry of uninformed LI traders is marked by the green-lined area between the points $\beta = \mu_L$ and $\beta = \beta_E^L$.

We use [Figure 6](#) to summarize our three-step proof of [Proposition 3](#) parts (a) and (c). First, we show that under any regular pdf f , the marginal welfare $\Delta_W(\beta)$ is increasing over low informed ratio β and decreasing over high β . Intuitively, the marginal welfare Δ_W is small if either the hedging benefit of a marginal uninformed trader is small or if few uninformed traders remain at the margin. When the informed ratio β is sufficiently low, the bid-ask spread is near zero. Hence, the marginal uninformed trader's hedging benefit is small, which

implies that the marginal welfare Δ_W is small. When the informed ratio β is sufficiently high, the spread is wide and very few uninformed traders still trade, meaning the marginal welfare Δ_W is again small. Altogether, the marginal welfare Δ_W must be increasing from zero near the informed ratio $\beta = 0$ and decreasing to zero over a sufficiently high ratio β (regardless of whether Δ_W is n-shaped).

Second, we set the constants μ_L and μ_R such that marginal welfares $\Delta_W(\mu_L) = \Delta_W(\beta_E^L)$ and $\Delta_W(\mu_R) = \Delta_W(\beta_O^R)$, and show that closing the OTC market causes a net gain in welfare if the informed mass $\mu \leq \mu_L$ and a net loss if the mass $\mu \geq \mu_R$. From the left panel of [Figure 6](#), because the gross welfare gain is uniformly “taller” than the gross welfare loss, the area of gross welfare gain is larger than the area of loss. (The base lengths of the gain and the loss are equal when the label accuracy $\gamma = 1/2$.) This remains to be true if we shift the constant μ_L leftwards, and thus closing the OTC market generates a net welfare gain for all mass $\mu \leq \mu_L$. Similarly, on the right panel, the gross welfare loss is uniformly “taller” than the gross welfare gain, which would remain so as we shift the constant μ_R rightwards. Thus, closing the OTC market would cause a net welfare loss for all mass $\mu \geq \mu_R$. We do not require marginal welfare Δ_W be n-shaped in this argument to establish part (a).

Third, the net welfare gain is increasing in the mass of informed traders $\mu \in [\mu_L, \mu_R]$ if marginal welfare Δ_W is n-shaped. From both panels of [Figure A.3](#), when the informed mass μ moves from the lower constant μ_L towards the higher constant μ_R , the derivative of the net welfare change with respect to μ , written geometrically, is

$$\underbrace{\left(\theta \cdot \|\overline{CD}\| - \frac{1}{2} \cdot \|\overline{EF}\| \right)}_{\text{Derivative of the gross welfare gain}} - \underbrace{\left(\frac{1}{2} \cdot \|\overline{EF}\| - (1 - \theta) \cdot \|\overline{AB}\| \right)}_{\text{Derivative of the gross welfare loss}}.$$

The derivative is negative for any informed mass μ between μ_L and μ_R , as both segments $\overline{AB}$ and $\overline{CD}$ are shorter than $\overline{EF}$. Thus, increasing the informed mass μ from μ_L to μ_R monotonically reduces the net change in welfare, which ensures a unique cutoff in (μ_L, μ_R) at which the net change in welfare crosses 0 from above. Part (c) follows.

In the complete proof in [Appendix A.2](#), we let the label accuracy γ be any value, not only $1/2$. For any accuracy $\gamma \neq 1/2$, the base lengths of the losses and the gains in [Figure 6](#)

differ from each another. It happens that, in calculating welfare, the relative weights on the gross loss and the gain change inversely to keep their base-times-weight equal. Therefore, our comparison of the lengths $\{||\overline{AB}||, ||\overline{CD}||, ||\overline{EF}||\}$ continues to work in establishing Parts (a) and (c) for all values of label accuracy γ and θ , $\theta > 1 - \gamma$.

How general are the conditions in [Proposition 3](#)? Most economically relevant distributions would result in an n-shaped marginal welfare Δ_W . [Proposition 4](#) states the necessary and sufficient conditions on the distribution F —a model primitive—for the resulting marginal welfare Δ_W to be n-shaped and marginal volume Δ_V to be decreasing.

Proposition 4. (i) *Marginal welfare Δ_W is n-shaped if and only if the function*

$$\frac{(2\alpha - 1)(1 - F(x))}{xf(x)(2\alpha - 1 - x)^2} - \frac{1}{2\alpha - 1 - x} \quad \text{is U-shaped in } x \in (0, 2\alpha - 1). \quad (9)$$

In particular, any beta distribution $\text{Beta}(a, b)$ satisfies condition (9) for all $a, b > 0$.

(ii) *Marginal volume Δ_V is decreasing if and only if the function*

$$\frac{(2\alpha - 1)(1 - F(x))}{f(x)(2\alpha - 1 - x)^2} - \frac{x}{2\alpha - 1 - x} \quad \text{is increasing in } x \in (0, 2\alpha - 1). \quad (10)$$

Any beta distribution $\text{Beta}(a, b)$ with parameters $a \leq 1, b \leq 1$ satisfies condition (10).

Using Taylor expansion around $2\alpha - 1$, one can verify that the uniform distribution $\mathbb{U}([0, 1])$ and the beta distributions—for all parameter values—satisfy condition (9) and thus have n-shaped marginal welfare Δ_W .¹⁸ To interpret condition (9), we note that it holds at the extremes under *any* distribution F : the expression goes to infinity as x approaches 0 or $2\alpha - 1$. Condition (9) only rules out the distributions whose pdf f oscillates over moderate values in the support $(0, 2\alpha - 1)$.

We can verify that (10) is satisfied for any uniform distribution and for beta distributions $\text{Beta}(a, b)$ with parameters $a \leq 1, b \leq 1$. Such distributions starkly demonstrate the effect of cheap substitution. Because the entrants' hedging benefits are uniformly larger than the

¹⁸The beta distribution $\text{Beta}(a, b)$ (pdf $f(x) = \frac{x^{a-1}(1-x)^{b-1}}{B(a, b)}$) is a highly general class of bounded distributions that embeds the uniform distribution when $a = b = 1$. One can also numerically verify that common unbounded distributions—Normal, Chi-squared, and Gamma—satisfy condition (9) when truncated on $[0, 1]$.

exiters' benefits, closing the OTC market can raise welfare—even as it reduces aggregate volume—for the assets that attract relatively few informed traders.

4 Optimal Pigouvian Tax

Our focus so far on closing the OTC market has two benefits. First, the focus reflects the aims of recent policies. Second, we can provide clear guidance for the many assets that are predominantly traded over the counter (close the OTC market) or on exchanges (keep the OTC market). However, closing the OTC market is a crude policy without clear guidance for those assets whose OTC market shares are moderate.

In this section, we introduce a Pigouvian tax on the OTC market which *consistently* obtains a strictly higher welfare than closing the OTC market. The tax subsidizes trades on the exchange with the revenue levied from the OTC market. The OTC market share smoothly declines as the tax t increases.

Policymakers can easily implement the socially optimal tax. We identify a local experiment to determine if the current tax is too low, too high, or optimal.

4.1 Characterizing the Optimal Pigouvian Tax

We introduce a Pigouvian tax on the dealer to the model of [Section 2](#). Gross tax revenue t is transferred to the market maker on the exchange. The tax t is common knowledge from the beginning of the game. Our analyses fix all parameters $\{\mu, \theta, \gamma, \alpha\}$ except the tax t .

All implementations of a Pigouvian tax levied on the dealer that raise the same gross revenue t are equivalent to each other, as they all lead to the same zero-profit conditions.¹⁹ Precisely, the dealer's condition becomes

$$S_O(t) \cdot \underbrace{(1 - S_O(t) + \beta_O)}_{V_O(t)} \gamma = (2\alpha - 1)\beta_O \gamma + t, \quad (11)$$

¹⁹Examples include flat per-trade transaction tax, lump-sum tax, and non-linear levy on OTC volume.

and the market maker's becomes

$$S_E(t) \cdot \underbrace{(1 - S_E(t) + \beta_E)}_{V_E(t)} \cdot (1 - \gamma) = (2\alpha - 1)\beta_E \cdot (1 - \gamma) - t. \quad (12)$$

Increasing the tax t raises the OTC spread $S_O(t)$ and lowers the exchange spread $S_E(t)$.

We define the *optimal tax* as follows.

The *optimal Pigouvian tax* t^* is the tax t that maximizes welfare W . (13)

Proposition 5 says that imposing the optimal tax t^* attains a strictly higher welfare than closing the OTC market.

Proposition 5. *Given any triple (α, θ, γ) : (a) There exists a unique optimal Pigouvian tax t^* (defined in (13)). (b) Imposing the optimal tax t^* attains a strictly higher welfare W than closing the OTC market.*

The optimal Pigouvian tax *always* outperforms closing the OTC market. This is because cheap substitution vanishes under the extreme tax that effectively closes the OTC market. We define this *Zero-OTC* tax $\bar{t}$ as follows:

Zero-OTC tax $\bar{t}$ is the smallest tax such that the OTC trade volume is zero.

At the Zero-OTC tax $\bar{t}$, each trader is indifferent between trading over the counter or on the exchange. In particular,

$$S_O(\bar{t}) = S_E(\bar{t}).$$

Thereby the Zero-OTC tax equalizes the hedging benefits of the marginal uninformed traders in both markets. A marginal tax cut from the Zero-OTC tax $\bar{t}$ would create entrants and exiters with the same hedging benefits, eliminating cheap substitution.

Our proof shows that the optimal tax t^* is strictly below the Zero-OTC tax, $t^* < \bar{t}$ ([Appendix A.3](#)). Lower OTC tax generates welfare gain among the uninformed LU traders, while the lower subsidy on the exchange destroys welfare among the uninformed LI traders.

However, the informed traders partly absorb the tax and subsidy cut, and more so where the ratio of informed to uninformed traders β is higher. Overall, the informed LI traders absorb a higher proportion of the tax change than the informed LU traders, which raises welfare. Then, the optimal tax t^* must be strictly below the Zero-OTC tax $\bar{t}$, and the optimal tax must obtain a strictly higher welfare than closing the OTC market.

4.2 Implementing the Optimal Pigouvian Tax

We use the trade-off between the cheap substitution versus the volume effect (defined in (4)) to develop a simple procedure to implement the optimal Pigouvian tax t^* . Marginally increasing the tax t raises welfare W if and only if

$$\underbrace{\left| \frac{S_E}{S_O} \right|}_{\text{Cheap Substitution}} > \underbrace{\left| \frac{dV_O}{dV_E} \right|}_{\text{Volume Effect}}.$$

The spread ratio S_E/S_O captures cheap substitution, which raises welfare upon the tax increase. It is greater than 1, because the marginal uninformed traders have greater hedging benefits on the exchange than over the counter ($S_E > S_O$). By contrast, the volume effect $|dV_O/dV_E|$ captures the decline in trade volume. Welfare rises if and only if cheap substitution exceeds the volume effect, $|S_E/S_O| > |dV_O/dV_E|$.

We define the ratio of cheap substitution to the volume effect as the *weighted spread ratio*:

$$\text{WSR}(t) := \left| \frac{S_E(t)dV_E(t)}{S_O(t)dV_O(t)} \right|. \quad (14)$$

If the ratio $\text{WSR}(t) > 1$, the current tax t is too low relative to the socially optimal tax, $t < t^*$. If the ratio $\text{WSR}(t) < 1$, the tax t is too high, $t > t^*$.

Proposition 6. *We suppose the current Pigouvian tax is t and it marginally increases by a small amount $dt > 0$. (a) The consequent weighted spread ratio $\text{WSR}(t) > 1$ (defined in (14)) if and only if the current tax is strictly lower than the optimal tax $t < t^*$, and $\text{WSR}(t) < 1$ if and only if $t > t^*$. (b) Welfare W increases $dW/dt > 0$ if and only if $\text{WSR}(t) > 1$, and*

$dW/dt < 0$ if and only if $WSR(t) < 1$. (c) $WSR(t)$ is strictly decreasing in t .

Proposition 6 spells out a simple procedure to implement the optimal Pigouvian tax:

Procedure

1. Introduce a tax near zero.
2. Calculate the weighted spread ratio WSR .
3. If WSR is above 1, raise the tax; if WSR is below 1, lower the tax; otherwise, do nothing and end this procedure.
4. Repeat steps 2 and 3 until the procedure ends.

Estimating the Weighted Spread Ratio

The volume term of WSR $|dV_O/dV_E|$ is easily computed using trade volumes just before and just after the change in the tax t . Already in the US and the EU, most trades must be reported in nearly real time, specifying whether it was executed on an exchange or over the counter. The spread term S_E/S_O is empirically the ratio of the average bid-ask spread on exchanges to the average spread over the counter just before the tax change. To estimate it, the exchange spread S_E can be the best quoted ask minus the best quoted bid, divided by the midpoint price. If no exchange has a bid or an ask, the effective spread from recent exchange trades can substitute for the quoted spread. In the OTC market, one can approximate the spread S_O with the effective spreads of similar assets (e.g., corporate bonds at the same firm with similar maturity) or adopt benchmarks widely used among traders (e.g., MarketAxess' Composite+ or Bloomberg's BVAL).

Applicability of the optimal Pigouvian tax

The Pigouvian tax approach applies to assets that have some pre-trade anonymous methods of trading. Such methods include limit order books, batch auctions, dark pools, and all-to-all requests for quote we describe in [Section 5.2](#). Most important asset classes are in fact actively traded via these methods: equities, their futures and options, Treasuries, corporate bonds, and repurchase agreements (repos). [Footnote 1](#) lists examples.

5 Policy Implications

This section applies our results to recent policy debates.

5.1 Regulatory Efforts to Close OTC Markets

Recent regulations in the US, EU, and Japan seek to push OTC trades onto exchanges. One might be tempted to exempt mostly OTC-traded assets, as the large OTC market share reflects a preference for OTC trading. However, we find that closing the OTC market would improve utilitarian welfare for precisely those OTC-dominated assets ([Proposition 2](#)).

Case in point, the US Dodd-Frank Act sought to push the trades of swaps onto exchanges by ensuring access to exchanges. The Act establishes Swap Execution Facilities (SEFs), a form of automated trading platform, and requires most swaps trades to be initiated on a SEF. Each SEF must offer a limit order book, and OTC dealers cannot own more than 40% of any SEF. In addition, a “fair access” rule restricts the conditions or fees that the SEFs can impose on their limit order books. As our model would predict, the access-based approach of Dodd-Frank failed to dislodge the OTC dominance in swaps—an asset class that our results imply should be traded on exchanges.²⁰ Yet some 95% of swaps trades remain over the counter several years after the implementation of the Act ([Nagel, 2016](#)).

Unlike the Dodd-Frank Act, the EU’s Markets in Financial Instruments Directive II (MiFID II) bans price discrimination by dealers. Specifically, MiFID II requires all dealers to maintain public and binding quotes for any asset that meets a minimum liquidity requirement. All except large block trades must transact at those public quotes or on exchanges ([Surowiecki, 2018](#)). This mandate-based approach of MiFID II is more likely to shift trades onto exchanges than the access-based approach of the Dodd-Frank Act. Nonetheless, in our model, the optimal Pigouvian tax ([Section 4](#)) would improve welfare upon both the mandate and the access-based approaches.

²⁰In our model, if there is no exchange initially and then one is introduced, the Likely Informed traders shift onto the exchange. For the assets mostly traded by Likely Uninformed traders, this shift will be small. This might explain why the market share of exchanges in standardized swaps stayed close to zero after the Dodd-Frank Act forced platforms to open swaps exchanges. We thank an anonymous referee for this point.

5.2 Post-trade Name Give-Up

The US regulators adopted the mandate-based approach when they banned “post-trade name give-up” in 2020. Most swaps trades—including over 80% for index credit default swaps and 45% for single-name CDS (Nagel, 2016)—are executed on SEFs, which offer two ways to trade: request for quote (RFQ) or all to all (A2A). In an RFQ, a trader simultaneously submits a non-anonymous request to trade a stated quantity to multiple dealers. Because an RFQ is pre-trade non-anonymous, it leads to an OTC trade in our model. An A2A trade occurs either on a limit order book or through a first-price auction open to all traders, both of which are pre-trade anonymous. The A2A protocol thus represents the exchange in our model.

Most SEFs traditionally practiced post-trade name give-up (NGU), wherein parties to an A2A trade learn each other’s identities after the trade. Buy-side firms intensely opposed NGU, since the “information leakage associated with sharing [their] trading activity” undermines A2A trading (Citadel LLC, 2020). The dealers claim NGU helps tailor their pricing on requests for quote and increases the liquidity they provide (JPMorgan Chase & Co., 2018). Such comments are consistent with NGU making the traders’ labels more accurate (higher θ). They also cite research that find RFQs have smaller spreads and generate more volume than A2A to argue for the status quo with NGU (Securities Industry and Financial Markets Association, 2018). Another major pro-NGU argument is that traders are “free to choose” whether or not to trade on SEFs with NGU. Despite these claims, the Commodity Futures Trading Commission (CFTC) banned NGU in 2020.

Did the CFTC make the right call? Yes, according to our results. As swaps trades are OTC-dominated (Nagel, 2016; Augustin, Subrahmanyam, Tang and Wang, 2016), **Proposition 2** predicts the ban on NGU (lowering label accuracy θ) to raise welfare. Further, **Proposition 2** undercuts a central argument for keeping NGU—that NGU increases trade volume or reduces bid-ask spreads does *not* imply welfare gain.

5.3 Permissioned Blockchain

We predict that the current plans to apply blockchain to financial markets would undermine exchanges in favor of OTC markets. Blockchain is an electronic recordkeeping procedure for a network of members, called “nodes.” Each transaction between the nodes is broadcast to the other nodes in the network, and each node maintains a ledger of all such transactions. An algorithm periodically reconciles those ledgers, and the sequence of reconciled ledgers forms the ‘official’ record. Pseudonymity is crucial to generating trust in this record. Transactions on a blockchain include anonymized identifiers of the counterparties involved in the trade. Thereby, the reconciled record provides the complete trade history of every account on the blockchain. Since pseudonymity violates anti-money-laundering rules (Elwell, Murphy and Seitzinger, 2013), most proposed blockchains for financial markets fully reveal traders’ identities to their nodes.

Among the earliest such proposals is the Depository Trust & Clearing Corporation’s (DTCC) plan to move the ownership records of most credit derivatives onto a blockchain (Irrera, 2017).²¹ The DTCC envisions a “permissioned” blockchain in which 15 major dealers are the nodes—and no one else (DTCC, 2018). Soon thereafter, ICAP (the dominant interdealer broker) initiated a pilot of a permissioned blockchain for foreign exchange, and the DTCC proposed a permissioned blockchain for equities (DTCC, 2020).

All three proposals would reveal all traders’ transaction histories to the major dealers, allowing the dealers to better separate traders by their trading motives (higher label accuracy θ in our model). Then the dealers can cream skim a greater share of uninformed traders, which worsens adverse selection risk on the exchanges and widens their bid-ask spreads (Proposition 0). The three proposed blockchains would therefore undermine liquidity on the exchanges, contrary to the the policymakers’ goal to encourage trading on exchanges.

²¹DTCC is the dominant clearinghouse for most securities.

6 Conclusion

We show that closing the OTC market can improve utilitarian welfare, under the conservative setup of competitive pricing in the OTC market. In practice, search frictions and the dealers' market power hamper price competition in OTC markets. As OTC trading moves onto electronic platforms, such frictions are dissipating (Hendershott and Madhavan, 2015; O'Hara and Alex Zhou, 2021; Hau, Hoffmann, Langfield and Timmer, 2021). Price discrimination by the dealers remains a fundamental feature of OTC trading. We show that the price discrimination leads to the resilience of OTC trading against regulations that encourage exchange trading. However, for precisely the most OTC-traded assets, closing the OTC market would *raise* welfare.

While the competitive pricing assumption in the OTC market underlines the strength of our results, that prices are competitive on the exchange takes away from it. Our model misses out on some important sources of inefficiency on exchanges. For example, we do not consider how price impact (Vives, 2011) or sniping by fast traders (Budish, Cramton and Shim, 2015) might interact with the choice between trading over the counter or on the exchange. We view fixing the inefficiencies of exchanges as a separate question from whether the OTC market should be closed. Those inefficiencies can be addressed by improving the design of exchanges as several studies propose.²² Moreover, these inefficiencies of exchanges dissipate when more low-(adverse selection)-risk traders participate. Our theory predicts most traders of OTC-dominated assets to be low-risk. Hence, it suggests that closing the OTC markets for such assets would introduce minimal inefficiencies onto the exchanges.

Previous work show that price discovery in secondary markets affects corporate investment decisions (for example, Goldstein and Guembel, 2008). We leave for future research the analysis of price discovery in the presence of an exchange and an OTC market for two reasons. In our model, a focus on price discovery *within* each market is an uninteresting one—the higher the informed ratio β in the market, the better is its price discovery. On the

²²Malamud and Rostek (2017) and Chen and Duffie (2021) show that optimal market fragmentation can address price impact, and Budish et al. (2015) proposes frequent batch auctions to resolve sniping by fast traders.

other hand, analyzing *aggregate* price discovery would require a stance on exactly how the quotes and transaction prices in the two markets are incorporated into the aggregate price discovery measure. For instance, specific price disclosure rules would determine the relative availability and importance of trade prices in the OTC versus the exchange markets.

Appendix

A Proofs

A.1 Proofs for Section 2.3

Proof of Proposition 0. Part (a): It suffices to show that there exists at least one solution to the zero-profit condition (1) so that $S(\beta)$ is well-defined. We use Figure A.1, which plots the market maker's payoff $s \cdot (1 - s) - (2\alpha - 1 - s)^+ \beta$ over the spread s . (With a general distribution F as in Section 3.3, the payoff is $s \cdot [1 - F(s)] - (2\alpha - 1 - s)^+ \beta$ and the proof is identical otherwise.) The payoff curve is continuous. Her payoff is negative at $s = 0$, as she is adversely selected yet has no revenue. It is positive at $s = 2\alpha - 1$, as she breaks-even on the trades against the informed and profits on the uninformed. The Intermediate Value Theorem implies that there exists at least one solution to the zero-profit condition. The smallest solution $S(\beta)$ is thus well-defined.

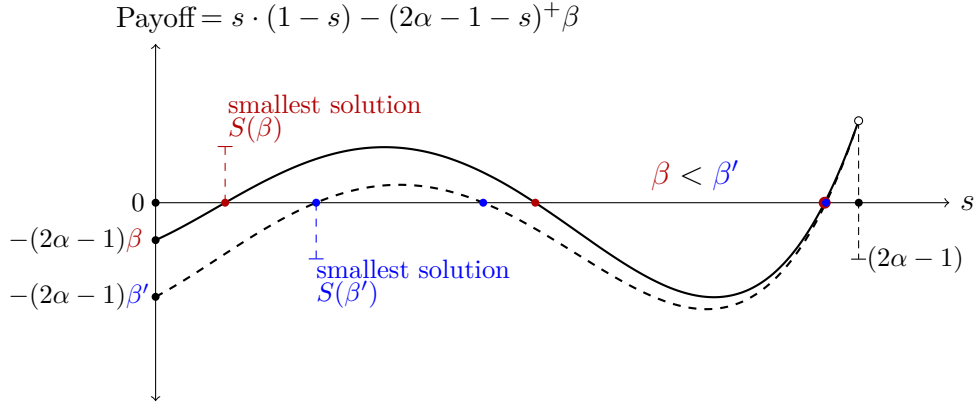


Figure A.1: Finding the equilibrium spread

Part (b): We proceed in three steps. First, we show that the spread function $S(\beta)$ is increasing in the informed ratio $\beta \in [0, \infty)$. We see this easily in Figure A.1: increasing β (to β') shifts the entire payoff curve downwards and the crossing point $S(\beta)$ to the right. Intuitively, as the informed traders impose losses on the market maker, he requires a wider spread to break-even when there are more informed traders.

Second, we solve for the spreads in the OTC market. All traders with the same label share the same OTC spread, because they are indistinguishable to the dealer. For an LU trader, the equilibrium OTC spread is $S(\beta_{\text{LU}})$, where β_{LU} is the informed ratio of the LU traders. As LU traders consist of $(1 - \theta)\mu$ informed and γ uninformed traders, their OTC spread is $S\left(\frac{1-\theta}{\gamma}\mu\right)$. Similarly, the LI traders' OTC spread is $S\left(\frac{\theta}{1-\gamma}\mu\right)$.

Third, we turn to the exchange spread S_E . If the market maker sets $S_E \in (S(\beta_{\text{LU}}), S(\beta_{\text{LI}})]$, all LU traders choose the OTC market, whereas all LI traders choose the exchange. Then the informed ratio on the exchange $\beta_E = \beta_{\text{LI}}$. The market maker thus earns zero profit if and only if she sets $S_E = S(\beta_{\text{LI}})$ in this case. If the market maker sets $S_E \leq S(\beta_{\text{LU}})$, then every trader chooses the exchange, implying $\beta_E = \mu > \beta_{\text{LU}}$, and thus the market maker earns a non-zero profit. Therefore, in equilibrium, (i) the exchange spread is $S_E = S(\beta_{\text{LI}}) = S\left(\frac{\theta}{1-\gamma}\mu\right)$, the lowest spread that earns the market maker a zero profit, and (ii) all LU traders choose the OTC market, whereas all LI traders choose the exchange. $\square$

Proof of Proposition 1. Part (a): The proof of Proposition 0 part (b) shows that the spread function $S(\beta)$ is strictly increasing in the informed ratio β . Since $\beta_{\text{O}} < \beta_{\text{E}}$, then $S_{\text{O}} < S_{\text{E}}$.

Part (b): The ratio $V_{\text{E}}/V_{\text{O}}$ equals

$$\frac{V_{\text{E}}}{V_{\text{O}}} := \frac{(1 - \gamma)(1 - S_{\text{E}}) + \theta\mu}{\gamma(1 - S_{\text{O}}) + (1 - \theta)\mu}.$$

The derivative of $V_{\text{E}}/V_{\text{O}}$ with respect to μ is positive if and only if

$$\frac{-\frac{\gamma}{1-\theta}\frac{\partial S_{\text{O}}}{\partial \mu} + 1}{\frac{\gamma}{1-\theta}(1 - S_{\text{O}}) + \mu} < \frac{-\frac{1-\gamma}{\theta}\frac{\partial S_{\text{E}}}{\partial \mu} + 1}{\frac{1-\gamma}{\theta}(1 - S_{\text{E}}) + \mu}. \quad (\text{A.1})$$

It suffices to show that

$$\frac{\gamma}{1-\theta}(1 - S_{\text{O}}) > \frac{1-\gamma}{\theta}(1 - S_{\text{E}}) > 0, \quad (\text{A.2})$$

$$\text{and } \frac{1-\gamma}{\theta}\frac{\partial S_{\text{E}}}{\partial \mu} < \frac{\gamma}{1-\theta}\frac{\partial S_{\text{O}}}{\partial \mu} < 1. \quad (\text{A.3})$$

Inequality (A.2) follows from $\gamma/(1 - \theta) > (1 - \gamma)/\theta > 0$ and $S_{\text{O}} < S_{\text{E}} < 2\alpha - 1 < 1$. One

can show that $S(\beta)$ is strictly concave. Then, (A.3) follows as

$$\begin{aligned} \frac{1-\gamma}{\theta} \frac{\partial S_E}{\partial \mu} &= S' \left(\frac{\theta}{1-\gamma} \mu \right) < S' \left(\frac{1-\theta}{\gamma} \mu \right) = \frac{\gamma}{1-\theta} \frac{\partial S_O}{\partial \mu} \\ &< S'(0) = 2\alpha - 1 < 1. \end{aligned} \quad \square$$

A.2 Proofs for Section 3

We proceed with the proofs in reverse order. We first prove [Proposition 3](#). Applying [Proposition 3](#) to the case $F = \mathbb{U}[0, 1]$ proves [Proposition 2](#).

The proof of [Proposition 0](#) shows that the spread function $S(\beta)$ is increasing. As $S(\beta)$ is also left-continuous in β , then $S(\beta)$ is left-differentiable. We let $S'(\beta)$ be the left derivative. Marginal volume Δ_V and marginal welfare Δ_W as in (6) and (7) are thus well-defined.

Proof of Proposition 3. Closing the OTC market is equivalent to lowering label accuracy θ from the current level $\theta_1 > 1 - \gamma$ to the uninformative level $\theta_0 := 1 - \gamma$. Hence, it suffices to establish [Proposition 3](#) parts (b)-(d) for when label accuracy θ falls.

Parts (b)-(d): The change in aggregate trade volume V as the label accuracy θ falls from θ_1 to θ_0 is

$$\underbrace{(1-\gamma) \int_{S(\frac{\theta_0}{1-\gamma}\mu)}^{S(\frac{\theta_1}{1-\gamma}\mu)} f(s) \, ds}_{\text{Entry by uninformed LI-traders}} - \underbrace{\gamma \int_{S(\frac{1-\theta_1}{\gamma}\mu)}^{S(\frac{1-\theta_0}{\gamma}\mu)} f(s) \, ds}_{\text{Exit by uninformed LU traders}},$$

which is equal to

$$(1-\gamma) \int_{\frac{\theta_0}{1-\gamma}\mu}^{\frac{\theta_1}{1-\gamma}\mu} \Delta_V(\beta) \, d\beta - \gamma \int_{\frac{1-\theta_1}{\gamma}\mu}^{\frac{1-\theta_0}{\gamma}\mu} \Delta_V(\beta) \, d\beta. \quad (\text{A.4})$$

Weighting (A.4) by the hedging benefit, we find the change in welfare to be

$$(1-\gamma) \int_{\frac{\theta_0}{1-\gamma}\mu}^{\frac{\theta_1}{1-\gamma}\mu} \Delta_W(\beta) \, d\beta - \gamma \int_{\frac{1-\theta_1}{\gamma}\mu}^{\frac{1-\theta_0}{\gamma}\mu} \Delta_W(\beta) \, d\beta. \quad (\text{A.5})$$

The proofs are intuitive with the aid of graphs. [Figure A.2](#) plots a generic Δ_W . We first

show that Δ_W begins at $\Delta_W(0) = 0$, from which Δ_W strictly increases before eventually strictly decreasing to zero. When $\beta = 0$, $S(0) = 0$, and so $\Delta_W(0) = 0$. As β becomes large, $S(\beta)$ approaches $2\alpha - 1$. Then,

$$S'(\beta) = \frac{1}{\beta'(S(\beta))} = \frac{2\alpha - 1 - S(\beta)}{\frac{(2\alpha-1)[1-F(S(\beta))]}{2\alpha-1-S(\beta)} - S(\beta)f(S(\beta))} \xrightarrow{\beta \rightarrow \infty} 0,$$

and thus $\Delta_W(\beta) \rightarrow 0$ as $\beta \rightarrow \infty$. One can verify that $\ln(\Delta_W)$ is differentiable in a neighborhood of $\beta = 0$ and

$$(\ln \Delta_W)'(\beta) = (\ln(S'))'(\beta) + S'(\beta) \left(\frac{1}{S(\beta)} + \frac{f'(S(\beta))}{f(S(\beta))} \right) \xrightarrow{\beta \downarrow 0} \infty.$$

Thus, $\ln(\Delta_W)$ is strictly increasing in a neighborhood of $\beta = 0$. That is, there exists some $\beta^L > 0$ below which Δ_W is strictly increasing. By a similar argument, there exists some $\beta^H > \beta^L$, above which $\Delta_W(\beta)$ is strictly decreasing. Altogether, Δ_W begins at $\Delta_W(0) = 0$, from which Δ_W strictly increases before eventually strictly decreasing towards the lower limit of zero.

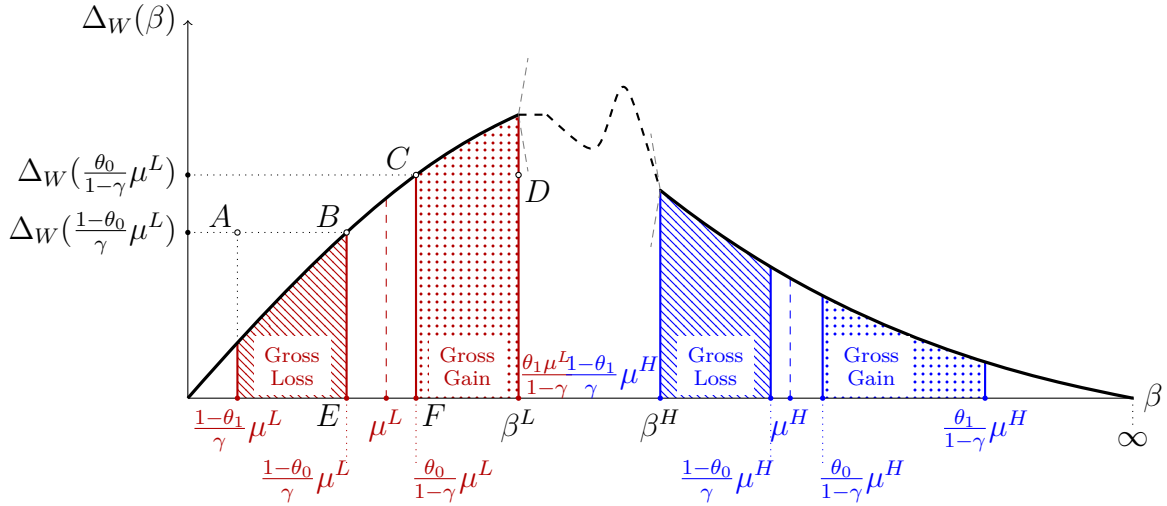


Figure A.2: Generic Δ_W

We define μ^L such that $\frac{\theta_1 \mu^L}{1-\gamma} = \beta^L$ (see Figure A.2). If $\mu = \mu^L$, the second term in (A.5)

(marked “Gross Loss” in red) has a strict upper bound

$$\gamma \cdot \left(\frac{1-\theta_0}{\gamma} \mu^L - \frac{1-\theta_1}{\gamma} \mu^L \right) \cdot \Delta_W \left(\frac{1-\theta_0}{\gamma} \right) = (\theta_1 - \theta_0) \cdot \mu^L \cdot \|\overline{BE}\| > 0,$$

which corresponds to the area ABE in [Figure A.2](#) scaled by the mass γ of uninformed LU traders. The first term in [\(A.5\)](#) (marked “Gross Gain” in red) has a strict lower bound

$$(\theta_1 - \theta_0) \cdot \mu^L \cdot \|\overline{CF}\|,$$

marked by the area CDF . Since the segment $\overline{CF}$ is longer than $\overline{BE}$ (because Δ_W is strictly increasing in $\beta \in [0, \beta^L]$), the Gross Gain in welfare is strictly larger than the Gross Loss. The same argument applies to any $\mu < \mu^L$, so that closing the OTC market raises welfare if the informed mass μ is small. Likewise, we choose μ^H such that $\frac{(1-\theta_1)\mu^H}{\gamma} = \beta^H$ and follow analogous steps to show that the Gross Loss in welfare (in blue) is larger than the Gross Gain if the informed mass μ is large $\mu \geq \mu^H$.

We next show that if Δ_W is n-shaped, there is a single cutoff on μ , denoted μ^* , above which a decrease in θ strictly reduces welfare and below which it strictly raises welfare. To prove this, we choose two constants, $\mu_R > \mu_L > 0$, as shown in [Figure A.3](#). We set μ_L to be the highest μ such that $\Delta_W \left(\left(\frac{(1-\theta_0)\mu}{\gamma} \right)^- \right) \leq \Delta_W \left(\left(\frac{\theta_1\mu}{1-\gamma} \right)^- \right)$, and set μ_R to be the highest μ such that $\Delta_W \left(\left(\frac{(1-\theta_1)\mu}{\gamma} \right)^- \right) \leq \Delta_W \left(\left(\frac{\theta_0\mu}{1-\gamma} \right)^- \right)$, where $\Delta_W(\beta^-)$ is the left limit of Δ_W at β . For ease of exposition only, [Figure A.3](#) illustrates the case where Δ_W is continuous in which case $\Delta_W \left(\frac{1-\theta_0}{\gamma} \mu_L \right) = \Delta_W \left(\frac{\theta_1}{1-\gamma} \mu_L \right)$ (line $\overline{BD}$) and $\Delta_W \left(\frac{1-\theta_1}{\gamma} \mu_R \right) = \Delta_W \left(\frac{\theta_0}{1-\gamma} \mu_R \right)$ (not shown). The same proof works if Δ_W is not continuous. As Δ_W is n-shaped, we know that such μ_L and μ_R exist, and that $\mu_L < \mu_R$. We proceed in two steps: We show that (i) the change in welfare [\(A.5\)](#) is strictly positive for all $\mu < \mu_L$ and strictly negative for all $\mu > \mu_R$; and (ii) [\(A.5\)](#) is strictly decreasing between μ_L and μ_R . Together, (i) and (ii) establish the existence of the single cutoff μ^* . For (i), we only show the case where $\mu \leq \mu_L$, since the argument is symmetric in the case where $\mu \geq \mu_R$. Setting $\mu = \mu_L$, an upper bound of the

second term in (A.5) (“Gross Loss” in Figure A.3) is

$$(\theta_1 - \theta_0) \cdot \mu_L \cdot \Delta_W \left(\left(\frac{(1 - \theta_0)\mu_L}{\gamma} \right)^- \right) = (\theta_1 - \theta_0) \cdot \mu_L \cdot ||\overline{BE}||$$

marked $ABEJ$ in Figure A.3. This upper bound is equal to the strict lower bound—marked $CDFH$ —on the first term of (A.5) (“Gross Gain”). Hence, (A.5) is strictly positive. To prove (ii) (A.5) is strictly decreasing over $\mu \in (\mu_L, \mu_R)$, the derivative of (A.5) with respect to μ , written geometrically, is

$$\underbrace{\left(\theta_1 \cdot ||\overline{DF}|| - \theta_0 \cdot ||\overline{GH}|| \right)}_{\text{Derivative of the gross welfare gain}} - \underbrace{\left((1 - \theta_0) \cdot ||\overline{BE}|| - (1 - \theta_1) \cdot ||\overline{IJ}|| \right)}_{\text{Derivative of the gross welfare loss}}. \quad (\text{A.6})$$

Due to Δ_W being n-shaped and how μ_L and μ_R are chosen, both $||\overline{BE}||$ and $||\overline{GH}||$ are strictly greater than $||\overline{DF}||$ and $||\overline{IJ}||$. Then, (A.6) is strictly negative. In sum, as θ decreases from θ_1 to θ_0 , the change in welfare is strictly positive if $\mu \leq \mu_L$, strictly negative if $\mu \geq \mu_R$, and strictly decreasing across $\mu \in (\mu_L, \mu_R)$, which together imply that a single cutoff μ^* exists.

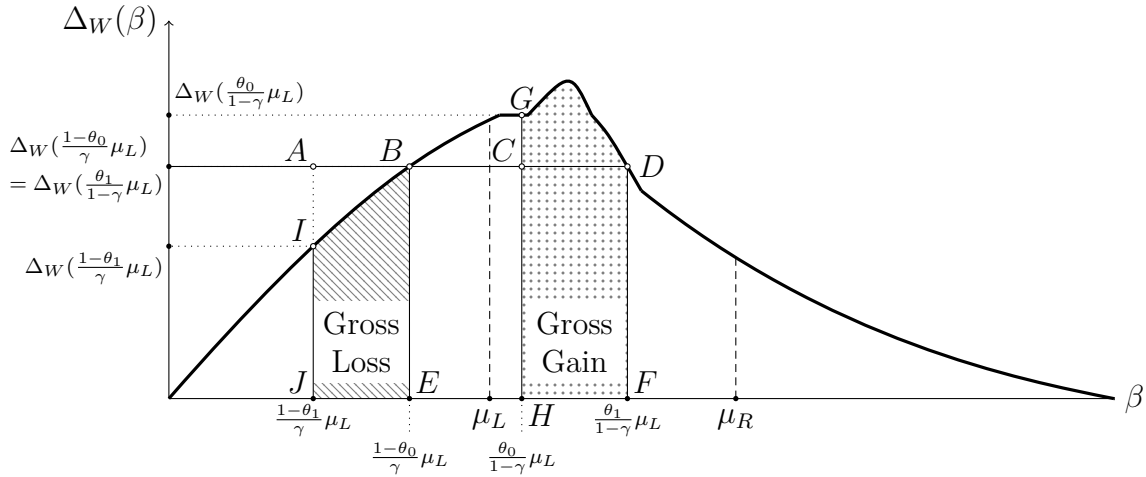


Figure A.3: “n-shaped” Δ_W

Lastly, we show that if Δ_V is strictly decreasing, the change in volume (A.4) is strictly negative. Figure A.4 plots a decreasing Δ_V . The second term of (A.4) (“Exiters” in Fig-

ure A.4) has a strict lower bound

$$(\theta_1 - \theta_0) \mu \cdot \|\overline{BE}\|,$$

marked ABE , which is larger than the first term's strict upper bound (“Entrants”)

$$(\theta_1 - \theta_0) \mu \cdot \|\overline{DF}\|,$$

marked CDF , and thus (A.4) is strictly negative. □

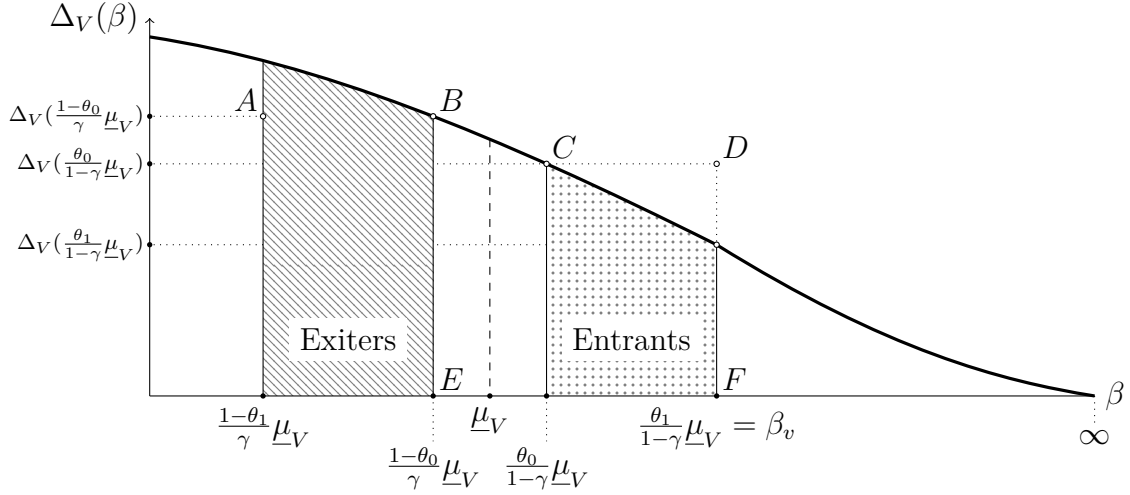


Figure A.4: Decreasing Δ_V

Proof of Proposition 4. The marginal welfare Δ_W can be written as

$$\Delta_W(\beta) = S'(\beta)S(\beta)f(S(\beta)) = \frac{S(\beta)f(S(\beta))}{\beta'(S(\beta))}.$$

Because the spread function $S(\beta)$ is strictly increasing in β , then $\Delta_W(\beta)$ is n-shaped if and only if $\beta'(x)/(xf(x))$ is U-shaped in $x \in (0, 2\alpha - 1)$. Differentiating (5) with respect to x yields

$$1 - F(x) + \beta - xf(x) = (2\alpha - 1 - x)\beta'(x),$$

which can be rearranged to

$$\beta'(x) = \frac{1 - F(x) + \beta - xf(x)}{2\alpha - 1 - x}.$$

From (5), we can express $\beta(x)$ as function of x :

$$\beta(x) = \frac{(1 - F(x))x}{2\alpha - 1 - x}.$$

Then, marginal welfare $\Delta_W(\beta)$ is n-shaped if and only if condition (9) holds. Likewise, marginal volume $\Delta_V(\beta)$ is decreasing if and only if (10) is true. $\square$

A.3 Proofs for Section 4

Proof of Propositions 5 and 6. We prove Propositions 5 and 6 together. First, imposing some Pigouvian tax t obtains a strictly higher welfare than closing the OTC market (Proposition 5 (b)). To show why, we proceed in three steps.

Step 1: We show that when $t = \bar{t}$, both the OTC and the exchange spreads are equal to the No-OTC spread S_N , $S_O(\bar{t}) = S_E(\bar{t}) = S_N$. The definition of $\bar{t}$ implies $S_O(\bar{t}) = S_E(\bar{t})$, wherein the dealer's and the market maker's zero-profit conditions are

$$\begin{aligned} S(\bar{t})(1 - S(\bar{t}) + \beta_O)\gamma &= (2\alpha - 1)\beta_O\gamma + \bar{t}, \\ S(\bar{t})(1 - S(\bar{t}) + \beta_E) \cdot (1 - \gamma) &= (2\alpha - 1)\beta_E \cdot (1 - \gamma) - \bar{t}. \end{aligned}$$

Summing the two equations gives

$$S(\bar{t})(1 - S(\bar{t}) + \mu) = (2\alpha - 1)\mu,$$

which shows $S_O(\bar{t}) = S_E(\bar{t}) = S_N$.

Step 2: We show that a marginal tax cut $-dt$ from the tax $\bar{t}$ increases the aggregate trade volume V . The volume V increases if and only if $|dV_O(\bar{t})/dV_E(\bar{t})| > 1$, where

$$dV_O(\bar{t}) = S'_O(\bar{t})\gamma dt$$

and

$$dV_E(\bar{t}) = -S'_E(\bar{t}) \cdot (1 - \gamma) dt.$$

Taking the derivative of the zero-profit conditions (11)-(12) with respect to t :

$$\begin{aligned} S'_O(t)(1 - 2S_O(t) + \beta_O)\gamma &= 1 \\ S'_E(t)(1 - 2S_E(t) + \beta_E) \cdot (1 - \gamma) &= -1. \end{aligned}$$

Then, using that $S_O(\bar{t}) = S_E(\bar{t}) = S_N$,

$$\left| \frac{dV_O(\bar{t})}{dV_E(\bar{t})} \right| = \left| \frac{S'_O(\bar{t})\gamma}{S'_E(\bar{t}) \cdot (1 - \gamma)} \right| = \frac{1 - 2S_N + \beta_E}{1 - 2S_N + \beta_O}.$$

The ratio $|dV_O(\bar{t})/dV_E(\bar{t})| > 1$ because $\beta_E > \beta_O$.

Step 3: We show that a strictly larger volume V implies a strictly higher welfare W . The entrants and the exiters upon a marginal change in the tax around $\bar{t}$ have the same hedging benefit. Since the entrants outnumber the exiters, welfare W is strictly higher.

Altogether, Steps 1-3 imply that the tax cut $-dt$ from $\bar{t}$ strictly raises welfare.

Second, we establish **Proposition 6** (b)-(c). Upon a marginal increase dt in the tax $t < \bar{t}$, the gross welfare loss among traders over the counter is $|S_O(t)dV_O(t)|$. The gross welfare gain among traders on the exchange is $|S_E(t)dV_E(t)|$. Thus, welfare W increases $dW/dt > 0$ if and only if $WSR(t) > 1$ (**Proposition 6** (b)). From the first half of this proof,

$$|S_O(t)dV_O(t)| = \frac{S_O(t)}{1 - 2S_O(t) + \beta_O} dt = \frac{1}{\frac{1+\beta_O}{S_O(t)} - 2} dt$$

and

$$|S_E(t)dV_E(t)| = \frac{S_E(t)}{1 - 2S_E(t) + \beta_E} dt = \frac{1}{\frac{1+\beta_E}{S_E(t)} - 2} dt.$$

Because $S_O(t)$ is strictly increasing in t while $S_E(t)$ is strictly decreasing, $|S_O(t)dV_O(t)|$ is

strictly increasing in t and $|S_E(t)dV_E(t)|$ is strictly decreasing. Hence,

$$\text{WSR}(t) = \left| \frac{S_E(t)dV_E(t)}{S_O(t)dV_O(t)} \right| = \frac{\frac{1+\beta_O}{S_O(t)} - 2}{\frac{1+\beta_E}{S_E(t)} - 2}$$

is strictly decreasing in t (**Proposition 6** (c)).

Therefore, there exists a unique $t^* \in [0, \bar{t})$ such that $\text{WSR}(t) > 1$ for $t \in [0, t^*)$ and $\text{WSR}(t) < 1$ for $t \in (t^*, \bar{t})$. **Proposition 6** (b) then implies that $dW/dt > 0$ for $t < t^*$ and $dW/dt < 0$ for $t > t^*$. That is, t^* is the unique optimal tax that maximizes welfare W . **Proposition 5** (a) and **Proposition 6** (a) follow. $\square$

Internet Appendices

I Request for Quote

We microfound the zero-profit condition and the trading protocol in the OTC market. We add two features to the model of [Section 2](#): There are $n \geq 2$ dealers and $m \geq 2$ market makers ($n + m$ intermediaries), and traders must request quotes from the dealers to trade over the counter. These additions do not change equilibrium allocation.

Strategies. In Stage 1, each trader chooses (i) whether to simultaneously request two-way quotes and (ii) from whom among the n dealers (possibly all n). (The equilibrium is the same whether the traders' choices (i) and (ii) are disclosed to the intermediaries or not.) In Stage 2, each market maker posts one bid and one ask available to all traders and, for each quote request received, each dealer offers a bid and an ask conditional on the requester's label. In Stage 3, each trader observes her prices then makes two decisions: (a) buy, sell, or exit and (b) against which price to trade. [Figure I.1](#) summarizes the timing and [Figure I.2](#) the traders' choices.

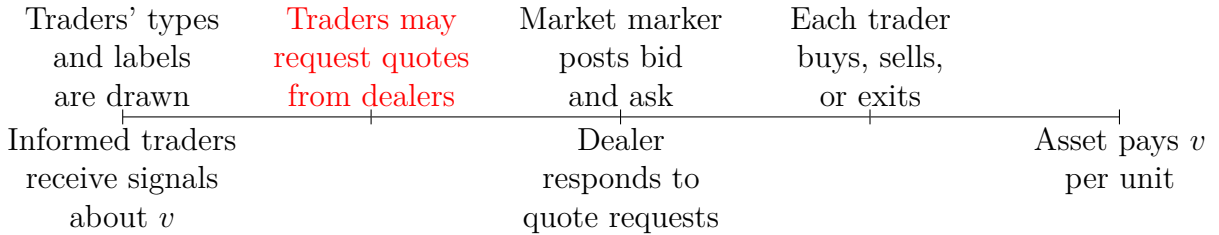


Figure I.1: Timeline

Equilibrium. We look for an extensive-form trembling-hand perfect (THP) equilibrium ([Selten, 1975](#)). The next proposition shows that the unique extensive-form THP equilibrium induces identical allocations as the equilibrium of the original model.

Proposition I.1. *There exists a unique extensive-form trembling-hand perfect equilibrium in which, with probability 1:*

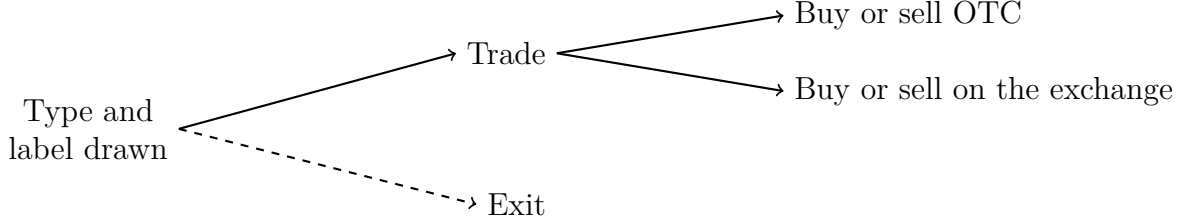


Figure I.2: Traders' Choices

- *Each trader sends a quote request to all the n dealers.*
- *Every LU trader chooses the OTC market and receives the OTC spread S_O , and every LI trader chooses the exchange and receives the exchange spread S_E . (The spreads S_O and S_E are given in [Proposition 0](#).)*

Proof. We first show that in Stage 1, each trader sends an RFQ to all the n dealers²³ with probability 1 under a THP equilibrium E . We fix a converging sequence of completely mixed strategy profiles E_n that gives rise to the equilibrium E . Each strategy profile E_n is a Nash equilibrium of a perturbed game. In Stage 1, each trader optimally submits an RFQ to every dealer with the maximum probability that is allowed in the perturbed game for two reasons. First, sending an RFQ is cost-free. Second, every dealer and market maker follows a completely mixed pricing strategy under the Nash equilibrium E_n , so any given dealer offers the overall best price with a strictly positive probability. Hence, each trader submits an RFQ to all the dealers with probability 1 under the limiting equilibrium E .

Since every trader submits an RFQ to all the dealers with probability 1, the extended trading game reduces to the original game with price posting instead of RFQ. Therefore, the extended trading game admits a unique THP equilibrium that induces identical equilibrium allocation as the equilibrium of [Section 2.3](#). $\square$

²³When it is costly for a dealer to respond to an RFQ, [Wang \(2022\)](#) shows that a trader submits an RFQ to only two dealers, and prices become noncompetitive. Adding such an OTC friction to our model would strengthen the benefit of closing the OTC market. We abstract away from any OTC friction to focus on the effect of cheap substitution.

II Evidence

OTC discount

A prediction of our model is that the bid-ask spread over the counter is lower than the spread on the exchange. Is this true? Recent evidence in [de Roure, Moench, Pelizzon and Schneider \(2021\)](#) suggests yes.²⁴ They examine data on German Bunds, which are traded both in the OTC market and on exchanges. Their headline result is that 87% of OTC trades execute at a better price (for the initiator of the trade) than the contemporaneous price on the exchange. A typical OTC discount is substantial: The median OTC trade saves 65.7% of the half spread it would have cost on the exchange. Moreover, the OTC discount is not an artifact of larger trade sizes, as [Seppi \(1990\)](#) would predict. The median trade size over the counter (EUR 5.4 million) is nearly identical to that on the exchange (EUR 5.0M). On average, the OTC trades are larger (EUR 12.74M vs 7.88M), solely because the biggest 5 percentile of trades are far larger over the counter (EUR 50M vs 13M). [Collin-Dufresne, Junge and Trolle \(2020\)](#) provides direct evidence of cream skimming: They find that dealers in index credit default swaps charge wider spreads to traders who seem to be informed.

Analogous evidence is found in the literature on the upstairs market for equities. (Upstairs trades are the OTC trades of stocks.) [Rose \(2014\)](#) compares upstairs and limit order book (LOB) trades on the Australian Stock Exchange. While OTC trades make a loss on average, LOB trades earn a profit. Further, traders who consistently make losses are more likely to trade upstairs and receive a discount relative to the price on the LOB. The opposite holds for consistently profit-earning traders. Others find that upstairs trades are less informed and less costly ([Madhavan and Cheng, 1997](#); [Smith, Turnbull and White, 2001](#); [Booth, Lin, Martikainen and Tse, 2002](#); [Bessembinder and Venkataraman, 2004](#); [Bernhardt, Dvoracek, Hughson and Werner, 2005](#); [Westerholm, 2009](#)). Large trades do not drive the upstairs discount. In fact, the discount is minimized among large orders ([Bernhardt et al., 2005](#)) and can be monotonically shrinking in trade size ([Westerholm, 2009](#)). The findings echo our mechanism that dealers offer a discount to the less informed traders and cream

²⁴We thank an anonymous referee for suggesting this paper.

skim them into the OTC market.

Migration of corporate bonds to the OTC market

Cream skinning in our model pulls a larger share of trades into the OTC market when a smaller share of traders are informed. The Depression-era migration of corporate bond trades is an episode consistent with this prediction. Most corporate bond trades moved from exchanges to the OTC market in the 1930s and 40s (Biais and Green, 2007). This movement coincided with a sudden, dramatic rise in the proportion of corporate bonds held by insurance companies and pension funds. Homer (1975) delivers the author’s first-hand account of this transition as a brokerage president in the 1930s: Pre-1930s, the corporate bond traders primarily traded on NYSE and were “small, country investors or big-city investors” (p. 379). Particularly active were sophisticated traders who “grabbed up” new issues, which “if well priced they sold at quick premiums” (p. 379). During the Great Depression, distressed investors sold out to dealers, who in turn resold the bonds to life insurance companies (p. 381). The life insurers preferred to trade over the counter although “[t]he exchange tried hard to retain its bond business” (p. 381). Because insurers typically trade with a hedging motive, the history of corporate bonds mirrors our prediction that OTC trading dominates where most traders are uninformed.

III Empirics

We use US equities data to document (1) substantial OTC market share in US-listed equities and (2) a positive correlation between the exchange market share and the exchange quoted spread, as predicted by Proposition 1.

III.1 Data, Variables, and Summary Statistics

We combine millisecond Trade and Quote (TAQ) data with weekly OTC trade volumes from the Financial Industry Regulatory Agency (FINRA).²⁵ In the FINRA data, OTC volumes are separated into Alternative Trading Systems (ATS) versus Non-ATS OTC volumes.

²⁵FINRA data used here is available publicly at <http://www.finra.org/industry/otc-transparency>.

The ATS consists of dark pools, batch auctions, and limit order books that are not designated as “national securities exchanges” by the US Securities and Exchange Commissions. Because trading on the ATS is anonymous, and thus does not allow trader-specific price discrimination, the ATS corresponds to the exchange in our model. The Non-ATS OTC refers to traditional bilateral and request-for-quote trades. Therefore, only the Non-ATS trades are counted as over the counter in our analysis.

The sample period is January 2, 2017–March 5, 2021, the available range of FINRA OTC data at the time of this analysis. We exclude all trades outside of market hours. Our sample consists of 3,210 US-listed non-ETF tickers that exist in both TAQ and FINRA data on both the first and the last weeks of the 218 weeks in the sample period. Only the trades during market hours are included to avoid an upward bias for OTC market share. (Results remain nearly identical if all trades are included).

Exchange market share for ticker i and week w is 1 minus the ratio of week w OTC dollar volume of trades from FINRA to the week w aggregate dollar volume from TAQ. To compute *percent quoted spread* for ticker i and week w , we use the millisecond TAQ quotes to calculate the time-weighted best quoted spread in percentage, $(\text{best ask} - \text{best bid}) / \text{midpoint}$, for each day, and then we take the simple average across the number of days observed for ticker i in week w . The total *number of trades* and total *dollar volume* of trades are computed for each ticker i and week w from millisecond TAQ trade data. [Table III.1](#) provides the summary statistics, first for all observations, and then by average weekly dollar volume of trades in quintiles.

III.2 Exchange Market Share for US-listed Equities

Does OTC trading dominate only for historically OTC-traded assets such as bonds? [Figure III.1](#) plots the average weekly exchange market shares of tickers in our sample. It shows that even among US-listed equities, many are OTC-dominated, and much of the remainder exhibit large OTC market shares.

Table III.1: Summary Statistics

Each observation is one ticker for one week.

All Observations								
	Obs	Mean	SD	Min	25%	50%	75%	Max
Exchange Market Share	694,305	0.750	0.198	0.000	0.597	0.828	0.914	1.000
Percent Quoted Spread	694,305	0.027	0.053	0.000	0.004	0.010	0.026	6.995
Dollar Volume of Trades	694,305	384.4M	2,752.6M	0.0M	3.3M	27.5M	187.2M	376,000.0M
Number of Trades	694,305	38,401	107,820	2	1,714	9,434	36,594	13,458,361
By Average Weekly Dollar Volume of Trades:								
Quintile 1 Observations								
	Obs	Mean	SD	Min	25%	50%	75%	Max
Exchange Market Share	137,455	0.615	0.185	0.000	0.484	0.609	0.758	1.000
Percent Quoted Spread	137,455	0.067	0.086	0.001	0.016	0.042	0.089	6.995
Dollar Volume of Trades	137,455	1.6M	3.7M	0.0M	0.4M	0.9M	1.9M	543.8M
Number of Trades	137,455	997	3,495	2	189	474	1,027	491,416
Quintile 2 Observations								
	Obs	Mean	SD	Min	25%	50%	75%	Max
Exchange Market Share	139,184	0.659	0.193	0.001	0.506	0.652	0.839	1.000
Percent Quoted Spread	139,184	0.037	0.055	0.001	0.011	0.022	0.044	4.943
Dollar Volume of Trades	139,184	8.3M	18.8M	0.0M	2.7M	5.4M	9.8M	1,570.1M
Number of Trades	139,184	5,240	14,348	7	1,345	2,815	5,745	950,784
Quintile 3 Observations								
	Obs	Mean	SD	Min	25%	50%	75%	Max
Exchange Market Share	139,010	0.775	0.194	0.005	0.667	0.859	0.922	1.000
Percent Quoted Spread	139,010	0.018	0.033	0.000	0.007	0.012	0.019	3.687
Dollar Volume of Trades	139,010	37.6M	50.9M	0.0M	16.2M	27.8M	46.3M	4,236.7M
Number of Trades	139,010	14,577	25,721	5	5,120	9,755	17,342	1,986,649
Quintile 4 Observations								
	Obs	Mean	SD	Min	25%	50%	75%	Max
Exchange Market Share	139,273	0.830	0.164	0.008	0.800	0.896	0.935	1.000
Percent Quoted Spread	139,273	0.008	0.012	0.000	0.003	0.005	0.009	0.487
Dollar Volume of Trades	139,273	153.5M	153.4M	0.0M	72.2M	119.7M	194.3M	14,100.0M
Number of Trades	139,273	32,783	36,820	59	14,249	25,524	41,738	3,485,211
Quintile 5 Observations								
	Obs	Mean	SD	Min	25%	50%	75%	Max
Exchange Market Share	139,383	0.867	0.108	0.013	0.851	0.902	0.931	1.000
Percent Quoted Spread	139,383	0.003	0.005	0.000	0.001	0.002	0.004	0.226
Dollar Volume of Trades	139,383	1,714.1M	5,957.4M	0.3M	427.1M	726.2M	1,409.3M	376,000.0M
Number of Trades	139,383	137,773	206,702	1,161	51,324	86,649	153,925	13,458,361

Table III.2: Summary Statistics by Exchange Market Share

Each observation is one ticker for one week. We present simple averages across observations with exchange market share less than 50% on the left column, and across observations with exchange share more than 50% on the right.

	Exchange Market Share < 50%	Exchange Market Share > 50%
Observations	80,746	613,559
Exchange Market Share	0.443	0.79
Percent Quoted Spread	0.031	0.026
Dollar Volume of Trades	40,283,547	429,702,482
Number of Trades	6,227	42,635

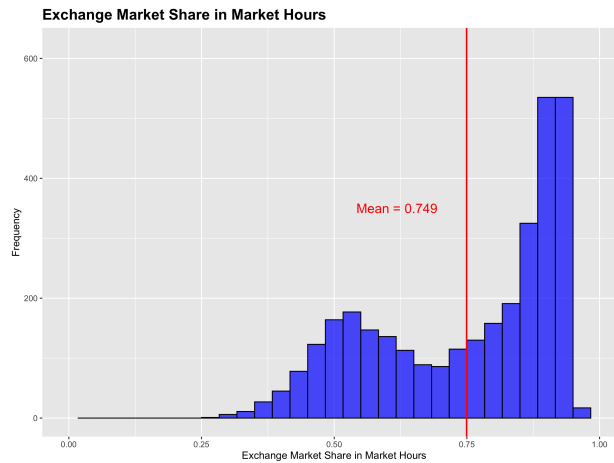


Figure III.1: Exchange Market Share of Dollar Value of Trades in US-listed Equities

III.3 Correlation between Exchange Market Share and Spread

Proposition 1 part (b) predicts a positive correlation between exchange market share and the exchange quoted spread *if* the observations vary substantively in the mass of informed traders μ but not in label precision (θ, γ) . We estimate the correlation within a narrow asset class, controlling for time fixed effects. Precisely, we partition US exchange-listed equities into quintiles by average weekly dollar volume and estimate the correlation within each quintile. Our empirical assumption is that across stocks within each dollar volume quintile in a given week, (i) the amount of available information about their individual traders (for example, the type of firm, reputation, past disclosures) is similar, so their label accuracies are also similar, and (ii) there is idiosyncratic variation in adverse selection risk (from, say, ticker-specific news).

Table III.3 presents the regression estimates for log exchange market share on log quoted spreads.²⁶ All regressions control for week fixed effects, thus our estimates capture cross-sectional variation. Standard errors are clustered at the ticker level. Under each quintile, the left-most regression has no controls. Consistent with **Proposition 1** (b), for every quintile, the correlation between log exchange market share and log quoted spread is positive. Our results do not depend on COVID or other time variation common across tickers, due to the week fixed effects.

A typical explanation for OTC trading is that certain assets are illiquid, which makes trading on exchanges impractical. We examine if our results are genuinely separate from the liquidity effect and, if so, whether the effect of our mechanism on exchange market share is in the same magnitude as the liquidity effect. To this end, the other regressions in **Table III.3** each control for one of two proxies for liquidity, namely, log weekly total dollar volume (“dollar volume”) or log weekly total number of trades. It shows that the quoted spread is positively correlated with the exchange market share independent of liquidity—if anything, the coefficient estimates for quoted spread become larger once we control for liquidity. That

²⁶Bogousslavsky and Collin-Dufresne (2022) reports a positive correlation between aggregate turnover and bid-ask spread. We examine the intensive margin of *where* trades occur and necessarily distinguish between the trades on exchanges versus over the counter. They examine the extensive margin of *how much* trading occurs in the aggregate.

Table III.3: Dependent Variable: $\log(\text{Exchange Market Share})$

All regressions control for week fixed effects. Observations are weekly and include 3,210 non-ETF US-listed tickers that exist in both the first and last weeks of the 218-week sample, from January 2, 2017 to March 5, 2021. Trades outside of market hours are excluded. Standard errors are clustered at the ticker level. Corresponding t-statistics are shown in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

Independent Variables	Quintile 1			Quintile 2			Quintile 3		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$\log(\text{percent quoted spread})$	0.074*** (11.61)	0.076*** (11.05)	0.072*** (11.66)	0.046*** (4.32)	0.144*** (15.57)	0.069*** (7.52)	0.061*** (3.62)	0.143*** (10.10)	0.077*** (5.71)
$\log(\text{dollar volume})$		0.005 (0.89)			0.129*** (24.39)			0.125*** (18.90)	
$\log(\text{number of trades})$			0.029*** (5.74)			0.123*** (19.27)			0.150*** (14.49)
R^2	0.059	0.059	0.068	0.027	0.168	0.182	0.029	0.140	0.208
N	137,453			139,184			139,010		
Independent Variables	Quintile 4			Quintile 5					
	(10)	(11)	(12)	(13)	(14)	(15)			
$\log(\text{percent quoted spread})$	0.087*** (5.74)	0.131*** (9.65)	0.095*** (8.16)	0.028*** (2.61)	0.038*** (2.98)	0.044*** (3.20)			
$\log(\text{dollar volume})$		0.095*** (10.30)			0.014 (1.57)				
$\log(\text{number of trades})$			0.146*** (13.08)			0.034*** (2.83)			
R^2	0.074	0.146	0.273	0.051	0.056	0.078			
N	139,273			139,383					

is, our findings are not explained by more liquid tickers being easier to trade on exchanges. Furthermore, the coefficient estimates for the quoted spread and the liquidity proxies are in the same magnitude. These findings suggest that our mechanism (of price discrimination by dealers) is not second-order to the typical illiquidity explanation.

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